ARPES viewer — User Guide

中文版 / Chinese version · *Version française*

Last updated: 28 October 2026 — for ARPES viewer v1.1.1

The version and its date are in the main window's title bar and at the bottom of the main window. Bug reports, questions and the latest version: https://github.com/DingPei1995/ARPES-data-browser (please give the version number when reporting a bug).

This guide walks through the program's main features with screenshots, in the order you would normally use them: loading data, looking at it, converting it to momentum, analysing it, and making figures. The technical background for each step (formulas, test results, design choices) is in the README and CHANGELOG.

Every screenshot was taken from the running program, using the synthetic data written by `devtools/demo_data.py`, so you can follow along without a beamtime (see Trying it with demo data).

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1. Installing and starting

You need Python 3.9 or newer and five packages:

```
pip install PyQt5 pyqtgraph numpy h5py scipy
```

Then, from the `ARPES_viewer` folder:

```
python ARPES_viewer.py
```

On a Mac you do not need Terminal to start it: double-click **`ARPES viewer.app`** in the program's folder (drag it to the Dock to keep it there). It finds the Python that has the packages above -- the same one as `python3` in Terminal, conda environments included -- and remembers it. If no Python has them, a dialog gives the `pip install` command (already copied to the clipboard: paste it into Terminal once) or lets you choose a Python yourself.

- If macOS says the app *cannot be opened because it is from an unidentified developer* (the program was downloaded as a ZIP): open **System Settings → Privacy & Security** and click **Open Anyway** (older macOS: right-click the app → **Open**). Only the first time.
- A copy of the app moved out of the program's folder (to Applications, say) asks once where the folder is.
- If the program stops with an error, the dialog shows the end of `~/.arpes_viewer/launcher.log`; please send that file with a bug report.

The program reads:

Beamline / source	Files
SOLEIL **ANTARES**	NeXus/HDF5 `*.nxs` (cuts, deflector maps, spatial scans)
SOLEIL **CASSIOPEE**	Scienta SES text files — single spectra, and numbered folders assembled into a map or a photon-energy scan
CASSIOPEE **spin end station** (MBS A-1)	`*.krx` and `*.txt` — cuts, deflector maps and spin-resolved EDCs
This program	its own saved `*.nxs` format

2. Trying it with demo data

```
python devtools/demo_data.py           # writes ./demo_data/*.nxs
```

This writes six files of synthetic data: a Fermi-surface map, a pair of cuts taken with LH and LV polarisation, a photon-energy scan, a spin-resolved EDC, a small real-space scan and a map with a detector mesh printed on it. They come from a toy band structure (a hexagonal electron pocket and a hole band at Γ, hole pockets at K) with a Fermi cut-off, the analyser resolution and Poisson noise, and open exactly like measured data.

To re-take every screenshot in this guide after changing a window:

```
python devtools/guide_screenshots.py    # writes docs/images/*.png
```

3. The main window



The main window is a **browser**: one row per *dataset*, not per file (one `.nxs` file often holds several measurements, and each gets its own row). The kind is written after the name: `[Cut]`, `[Map]`, `[kz map]`, `[k-map]`, `[SPEM]`, `[EDC]`, `[Spin EDC]` ...

It has three parts:

- **The tool panel** (left) — every analysis and processing tool, as a button, in two tabs. **Basic** (shown at start) holds what nearly every dataset goes through: Fermi edge fit, detector-grid removal, background subtraction, the three k conversions, the core-level check, aligning a kz map's Fermi levels and the Brillouin zone. **Advanced** holds the rest: figure composer, smoothing and the band-sharpening filters, symmetrising, stack plots, MDC and EDC fits, cut arithmetic, the slice series figure and the three 3-D views. Within a tab the tools are grouped by the data they take (*2D & 3D — cuts, maps*, *2D only — cuts*, *3D only — maps* ...).
- **The list** (right), with its buttons.
- **The Memory box** (bottom left) and the version line (bottom right).



The tool buttons follow the selection. A button lights up only when the rows selected in the list are what it takes — a k conversion (cut) needs one angle cut, cut arithmetic exactly two cuts, a 3-D view a map. A greyed button says why when you hover over it.

Action	How
Open a dataset in its own window	**double-click** the row (or right-click → **Open**)
Run a tool on the selected rows	its button in the **tool panel**
Stack several cuts into one map, or join a map / kz map measured in segments	**Combine...** under the list
Crop to a range, bin points, normalise each spectrum to itself	**Truncate...**, **Compress...**, **Self-normalise...** under the list
Open a map's slit or scan-axis cut	**right-click** the row (or a selection of rows)
Read the metadata, write the experiment log, rename, save, remove	the **Datasets** box of the tool panel, above Memory
See and free memory, close every viewer, open the autosave folder	the **Memory** box, bottom left
See which version you are running, report a bug, get the latest version	the title bar and the last line of the window (link to GitHub)

The four buttons under the list act on the rows selected above them, and each makes new rows — the originals are never changed.

The right-click menu holds only the ways of opening the rows: **Open**, **Open slit cut** and **Open scan-axis cut** (for a map). An entry that does not fit the selection is greyed, with the reason in its tooltip.

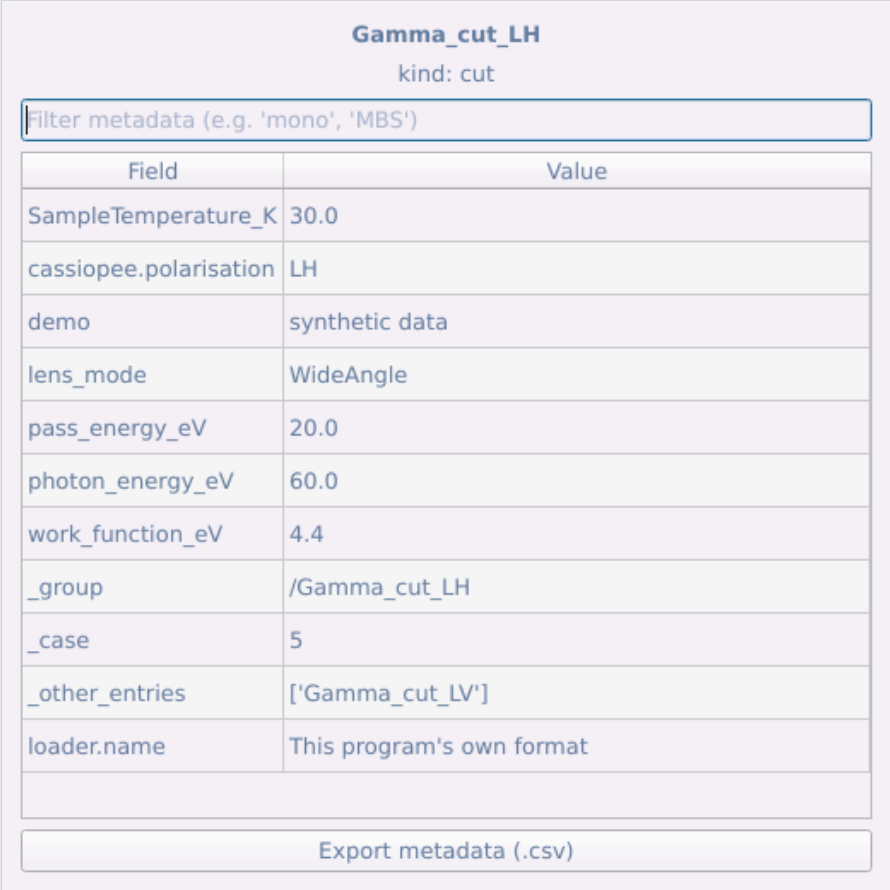
The Datasets box (tool panel, between the tools and Memory) holds the rest of what is done to the rows: **Experiment info**, **Log format...** and **Auto Exp. log** side by side (see the experiment log), **Rename...**, **Save datasets...**, **Remove from list** and **Session log...**. Its buttons grey out like the tools.

Where things are now (v1.1.1)

If you used an earlier version: the tools moved out of the right-click menu and the viewers' *Functions* menus into the tool panel, so each has one way in, always in sight.

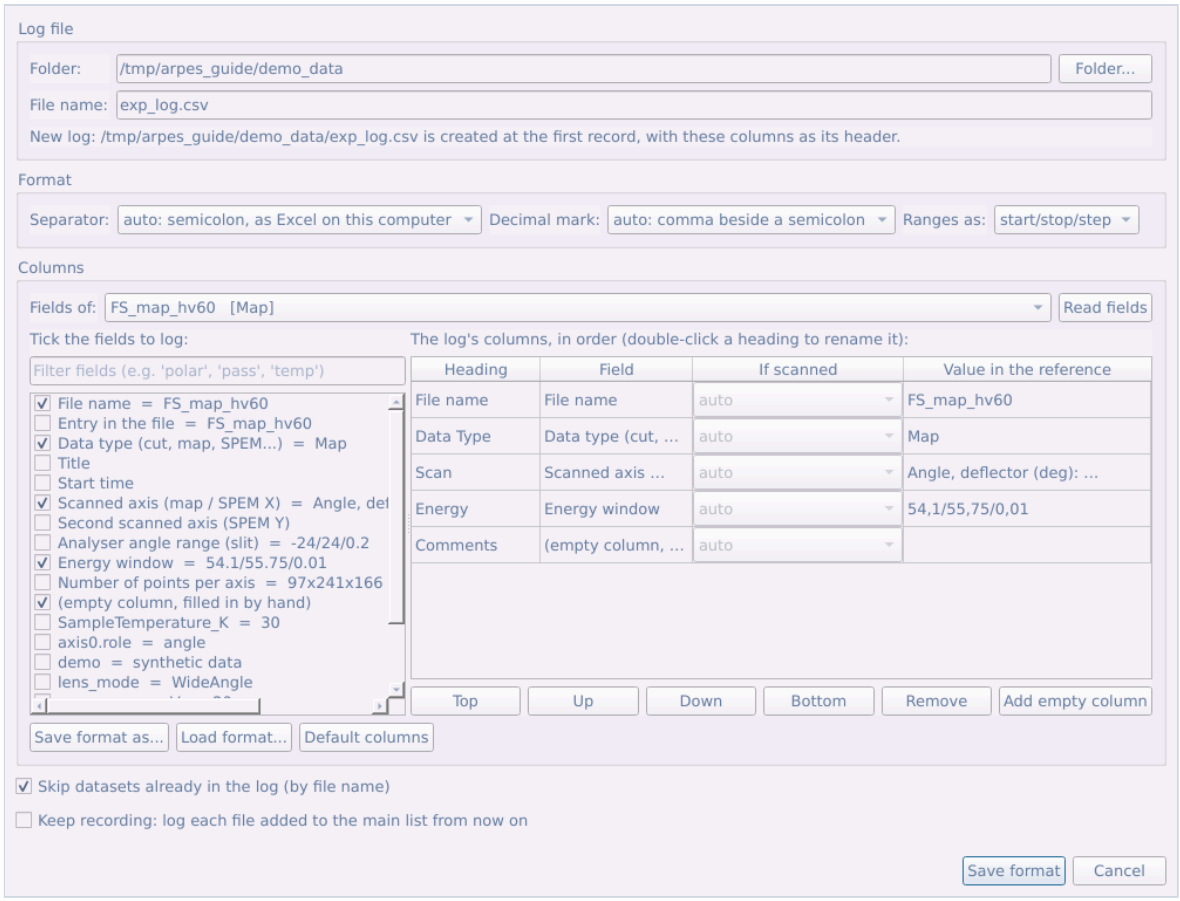
Now	Was
Tool panel → Basic → **Fermi edge fit...** (cuts and maps)	viewer *Functions → Fermi edge fit*
Tool panel → Basic → **Remove detector grid...**	viewer *Functions → Remove detector grid*
Tool panel → Basic → **k conversion (cut)** / **(map)**, **kz conversion**	viewer *Functions → Convert to momentum*
Tool panel → Basic → **Align Fermi levels (kz)...**	viewer *Functions → Align Fermi levels (kz map)*
Tool panel → Basic → **Brillouin zone...**	k-map viewer *Functions → Brillouin zone*
Tool panel → Basic → **Core-level check...**	right-click → Core-level check
Tool panel → Basic/Advanced → **Background subtraction**, **Smooth**, **Second derivative**, **Curvature**, **Symmetrise** (one window each)	**Smooth / derivative...** (one window with tabs)
Truncate..., **Compress...**, **Self-normalise...** under the list	**Crop / bin / normalise...**
Tool panel → Advanced → **MDC fit (k-cut only)**, **EDC fit**	viewer *Functions → MDC / EDC fit*
Tool panel → Advanced → **Cut arithmetic (two cuts)**	right-click / viewer *Functions → Cut arithmetic*
Tool panel → Advanced → **Slice series figure...**	map viewer *Functions → Multi-slice figure*
Tool panel → Advanced → **3D slices**, **Notched cube**, **Volume projection**	right-click → 3D view (one window, three modes)
Tool panel → Advanced → **Figure composer...**	right-click → Open in figure composer
Right-click → **Open scan-axis cut**, viewer button **Scan-axis cut**	Open deflector cut, Deflector cut
Memory box → **Free memory**, **Close all viewers**, **Open auto-save folder**	under the list
Datasets box → **Experiment info**	right-click → Show information
Datasets box → **Rename...**, **Save datasets...**, **Remove from list**, **Session log...**	right-click
Combine... under the list (only)	also right-click → Combine N datasets

Experiment info also opens the full metadata of the dataset in a window of its own, with a filter box and **Export metadata (.csv)**:



The experiment log

Two buttons side by side in the **Datasets** box write the beamtime's log -- a CSV file, one row per measurement -- from the files' own metadata: **Log format...** (left) sets it up once, **Auto Exp. log** (right) adds the selected datasets to it with one click.



Log format...

1. **Log file.** Choose the folder and type the file name. If a log of that name already exists, the window shows **that log's own format**, ready to adjust.

2. **Format.** *Separator* and *Decimal mark* are by default what Excel uses on this computer. Excel splits a CSV at the system's list separator -- a **semicolon** wherever the decimal mark is a comma (French, German, Italian... settings) -- so a comma-separated file opens there with everything in the first column. Leave them on *auto* unless the log is for a computer set up differently. *Ranges as:* writes a range `start/stop/step` or `start:step:stop`.

3. **Columns.** *Fields of:* picks a dataset of the main list -- choose one of the kind you log most -- and **Read fields** lists everything it has, with its value: the file name, data type, start time and scanned ranges worked out from the data, then every metadata field and motor. **Tick** a field to add it as a column. In the table on the right, double-click a heading to rename it (`hν/Δ`, `Polar/Δ`, `Epass`...), and use **Top** / **Up** / **Down** / **Bottom** to put the columns in any order. **Add empty column** gives one to fill in by hand (comments). **Save format as...** / **Load format...** keep one format per beamline.

4. **If scanned.** What a position or angle column says depends on the measurement:

   - a **cut**: the sample stood still -- the motor's value;
   - a **SPEM** scan: the scanned stage axis as `start/stop/step` (choose *SPEM: scanned X* / *Y* for the stage columns);
   - a **map** taken with the manipulator rather than the deflector: the angle as `start/stop/step` (*map: scanned angle*), and a sample position the beamline corrected along with the angle as `first..last`;
   - a **kz map**: the photon energy range (*kz map: photon energy*).

auto (the default) recognises the scanned motor by name, and a position that moved by its values. Any other measurement gets the field's own value.

5. **Save format.** If the log already holds records and you changed its format, it is **rewritten under the new header**: each record moves with its columns (renamed or moved columns are followed), new columns are empty for the old records, and columns you took out are named first, since what they hold goes. The header stays the only first row.

Skip datasets already in the log leaves out files whose name is already in the log's file-name column. **Keep recording** logs every file you add to the list from then on, until the program is closed: during a beamtime, add each file as it is measured and the log keeps itself.

Auto Exp. log adds the selected rows to the log, after its last record, without opening anything; the status bar says how many were added. The very first time, it opens *Log format...* first.

If the log is open in Excel, Windows will not let it be written: close it there and click again.

Combining datasets: stacking and joining

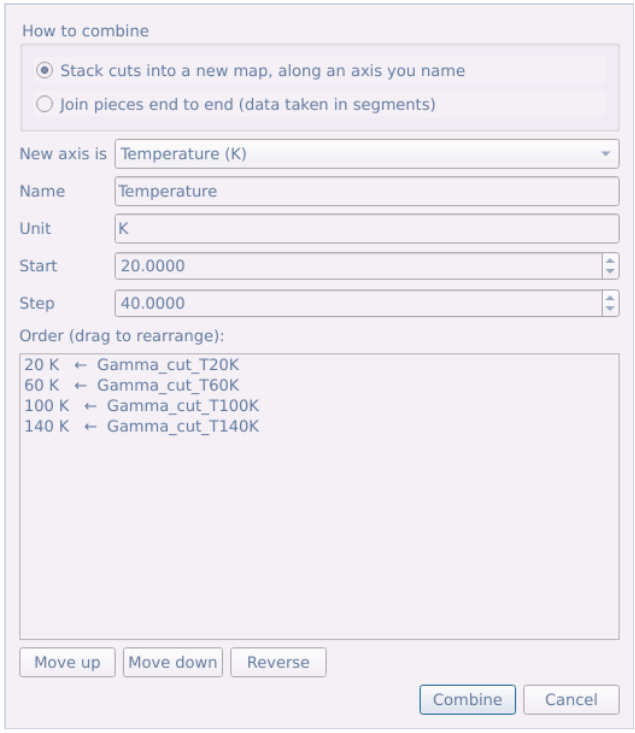
Combine... (under the list) does two different things, and asks which:

- **Stack** — *cuts only*. Each cut becomes one slice of a new map, along an axis you name: a temperature series, a gate-voltage series, one cut per photon energy. The result is looked at like any map — sliced at constant energy, or along the new axis.
- **Join** — *cuts, maps, kz maps or k-maps, all of one kind*. The pieces of one measurement that was taken in segments are put back end to end along one of their own axes: a map whose deflector range was scanned in two parts, a kz map whose photon-energy range was split over several runs, a cut whose energy window was taken in two halves.

Stacking cuts

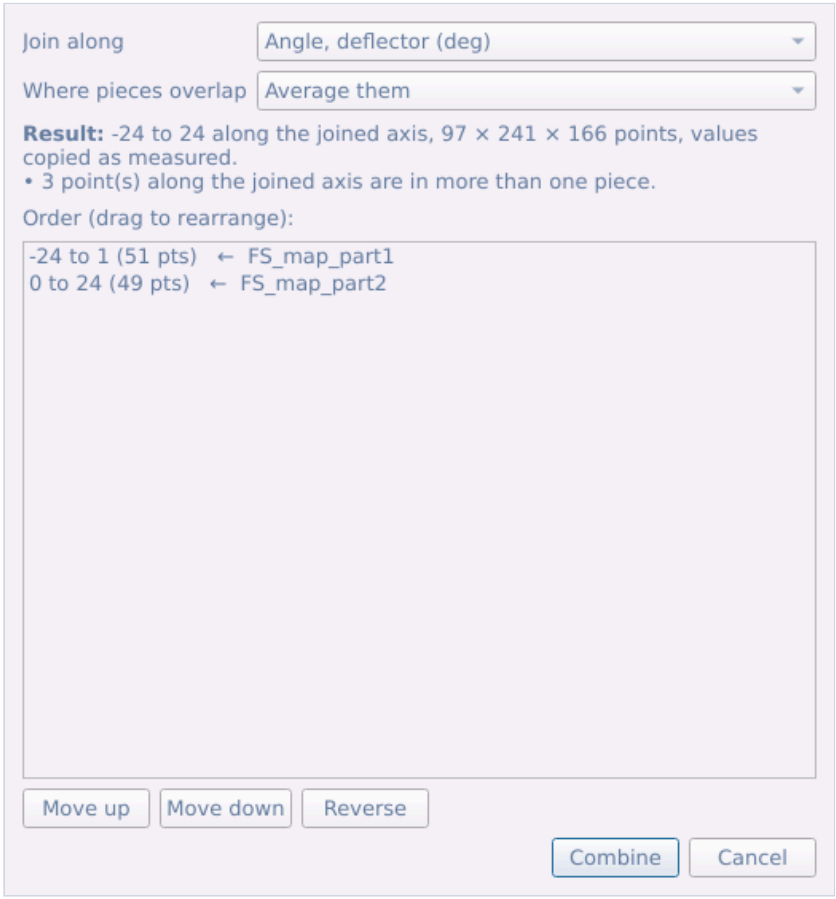
1. Select the cuts in the list (Ctrl/Shift-click) and press **Combine...**. Stacking needs the same axes point for point (same number of points, same values, same units); if they differ, *Stack* is greyed with the first difference named, and *Join* is offered instead.
2. It asks whether to combine them **in the list's order**. *Yes* keeps that order; *No* lets you choose it.
3. Choose **Stack**, then set the new axis: **New axis is** (temperature, gate voltage, photon energy, delay, position, other — this fills in *Name* and *Unit*, which you can change), **Start** and **Step**. Each cut is listed with the value it gets. If you chose to reorder, **drag** the rows (or use **Move up** / **Move down** / **Reverse**). Press **Combine**.

A photon-energy series of angle cuts becomes a `kz map`, so *Align Fermi levels (kz map)* and *Convert to momentum (kz, k∥)* take it (see §13); any other axis gives a `map` whose first axis is marked as not an angle, so no k conversion is offered for it.



Joining pieces of a measurement

1. Select the pieces (all maps, or all kz maps, or all cuts …) and press **Combine…**. Maps and kz maps go straight to *Join*.
2. **Join along**: the axis the measurement was split along. It starts on the axis whose range differs between the pieces — a map's deflector axis, a kz map's photon energy, or the energy of cuts taken in two energy windows — so it is usually right already.
3. **Where pieces overlap**: *Average them*, or keep the earlier / later piece in the order shown (drag the rows to change it).
4. The dialog lists each piece's range and shows the **Result** — the joined range and the number of points — with notes on anything it had to do. Press **Combine**.



How the axes are handled:

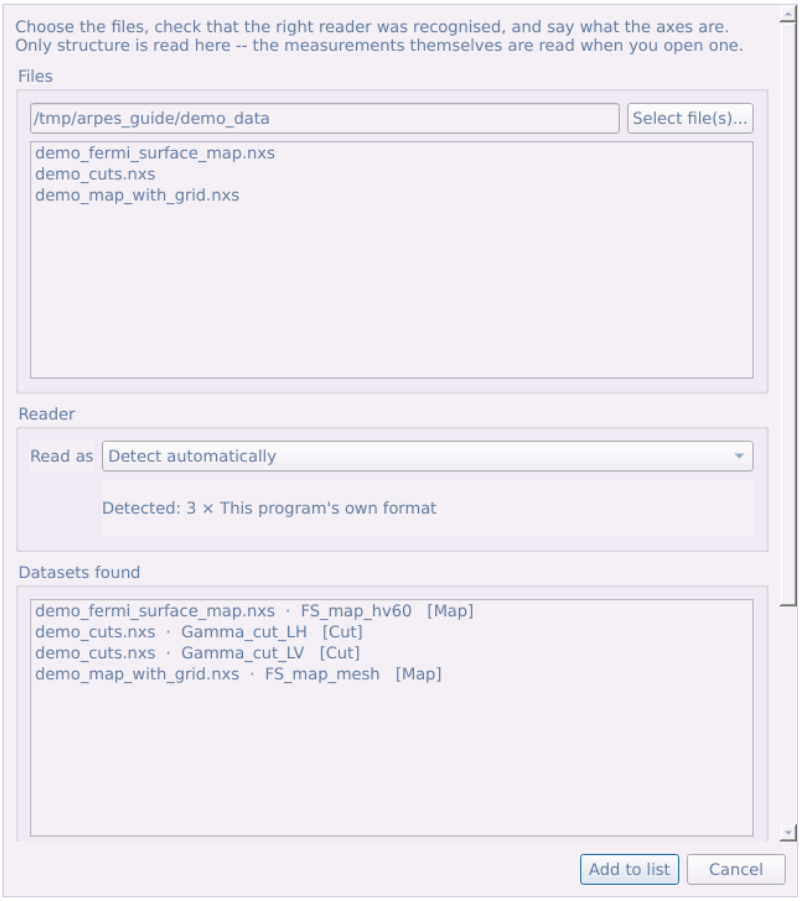
- Pieces on one common grid (same step, ranges that meet or overlap by whole points — the usual case) are **copied as measured**, nothing interpolated.

- A piece with a different step or a half-step offset is **interpolated** onto the common axis (the first piece's step), and the dialog says so.
- If the **other axes** differ — an energy window a few meV different between runs, a different number of slit channels — every piece is resampled onto the range they **all** cover, and the dialog says so.
- A stretch that no piece covers is a **gap**, left at zero and reported.
- A joined kz map is still a `kz map`: run *Align Fermi levels (kz map)* on the joined result rather than on the pieces (per-photon-energy fit results of the pieces are not carried over).

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4. Loading data

Load data... opens the loader. **Quick add...** skips it and adds the chosen files with everything detected automatically.



1. **Files** — pick one or many (*Select file(s)...*). The file filter is built from what the installed readers can open.
2. **Reader** — detected automatically, and the result is shown (*Detected: 3 × This program's own format*). Override it only if a file is being misread.
3. **Datasets found** — what is inside each file, read from its *structure* only. This is instant even for a multi-GB scan; nothing is read until you open a row.
4. **Axes** — *Axis order* reorders the array's dimensions once, at the door. *First axis of a map is* says what was scanned: a deflector angle (the default), a **photon energy**, a temperature, a gate voltage, a delay… This keeps the right unit everywhere afterwards, and stops a k conversion being offered for an axis that is not an angle.

CASSIOPEE folders. Select any file of a numbered series (`name_1_ROI1_.txt`, `name_2_ROI1_.txt`, …) and the whole folder is listed once, as one assembled cube. Whether it becomes an angle map or a kz map is read from the `_i` parameter files next to the spectra. The row is named with the photon energy and polarisation read from those files — `FineMap 117eV LH`, or `KzScan 40-120eV LH` for a photon-energy scan — so repeats of the same scan at another energy or polarisation can be told apart.

ANTARES photon-energy (PhD) files. A PhD `.nxs` file (one `Phd_NNNN` entry per photon energy) is listed as one `kz map`: the frames are sorted by photon energy and each kinetic-energy axis is referred to E_F with its own hν (work function 4.5 eV, which the k_z conversion then uses). Each frame still opens on its own as a cut.

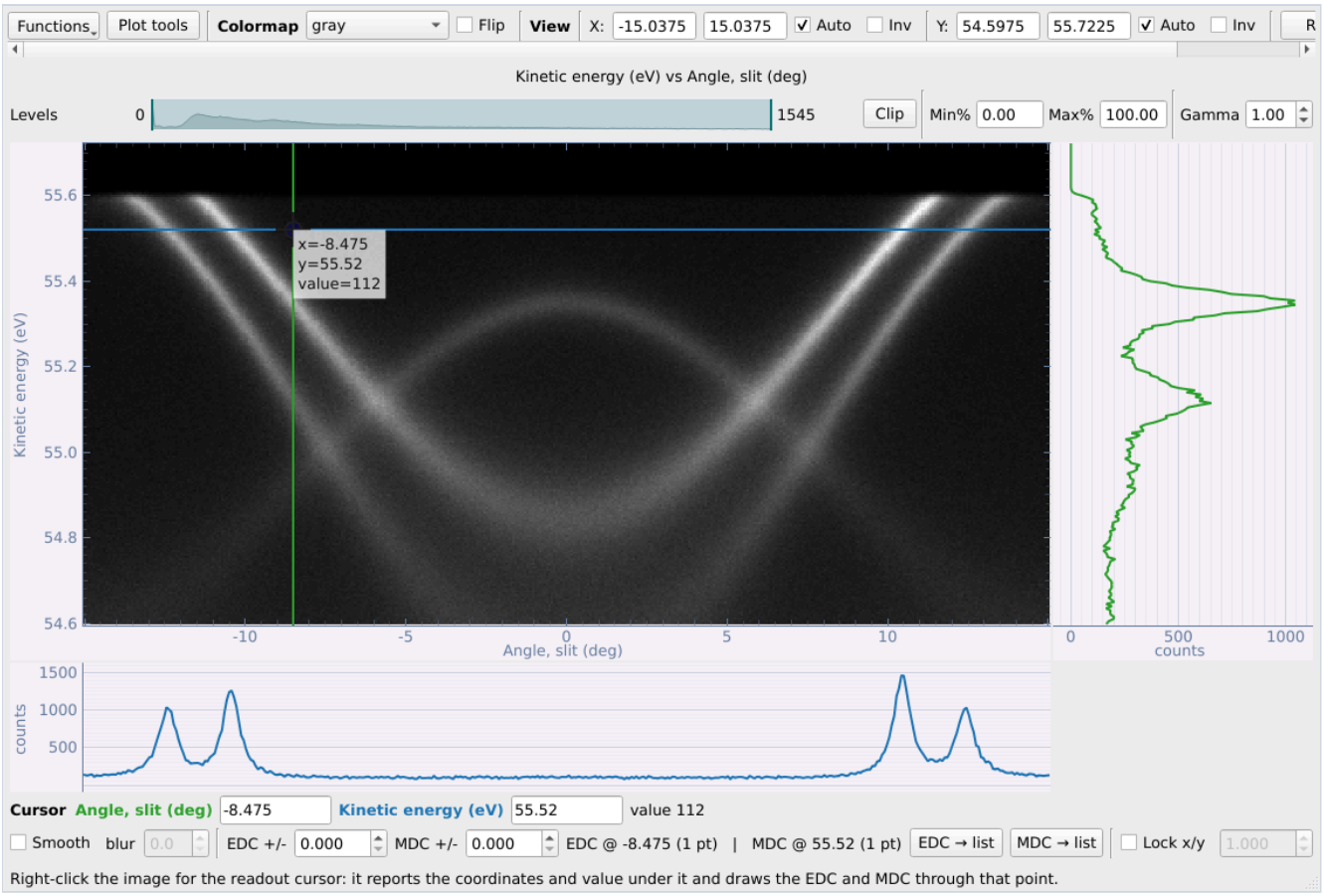
Data kinds

Kind	Axes	What it is
`cut`	angle (or k) × E	one spectrum
`map`	angle × angle × E	a deflector or polar-angle scan
`k_map`	kx × ky × E	a map converted to momentum
`kz_map`	hν × angle × E	a photon-energy scan
`kz_map_k`	k_z × k∥ × E	that scan converted to momentum
`spem_4d`, `spem_1d`	x (× y) × angle × E	real-space scans
`edc`, `mdc`, `spin_edc`	E (or k) × channels	curves, with optional σ channels

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5. What the viewers have in common

Each dataset opens in its own window, so you can put a map, its cuts and another file side by side.



- **The top row**: the **Functions** menu (everything the window can do), **Plot tools** (guide lines and labels on this plot — see below), the colormap (per window) with **Flip**, and the view controls — axis ranges, **Auto**, **Inv** (invert), **Reset**, **Grid**.
- **Above each image**: the **Levels** bar (drag the two handles, or type **Min%/Max%** percentiles; **Clip** applies them) and **Gamma**. A min/max you set is kept while you move the slice (the energy of a contour, the angle of a cut, the cursor of a spatial map), so every position is on the same colour scale; drag the handles back to both ends, or set 0%–100%, to follow each frame's own range again.
- **Under each image**: **Smooth** (display only, does not change the data) and **Lock x/y** (aspect ratio).
- **Readout cursor** — right-click the image → *Readout cursor*. It reports the value under it and, on a cut, draws the **EDC** (right) and **MDC** (below) through that point. **EDC ±** / **MDC ±** integrate over a window, shown as a shaded band on the image. Type a position into the **Cursor** row and press Enter to put it exactly there.
- **EDC → list** / **MDC → list** put the curve on screen in the main list as a dataset of its own (see §15).
- **Right-click the image → Export** saves what the panel shows — data (`.nxs`), an image, or vector axes (`.eps`); see §17.
- **Functions → Slice → Save slice to the main list / Freeze slice in a new window** keeps a copy of the slice on screen. The frozen window (and *Open this position in a new window* on a spatial map) opens with the panel's min/max and gamma, so the two can be compared directly. It has its own **Plot tools**, and a **Figure composer...** button to make a figure of it (§17).

The **Functions** menu now holds only what works on the window itself — the tools that take a dataset are in the launcher's tool panel (§3):

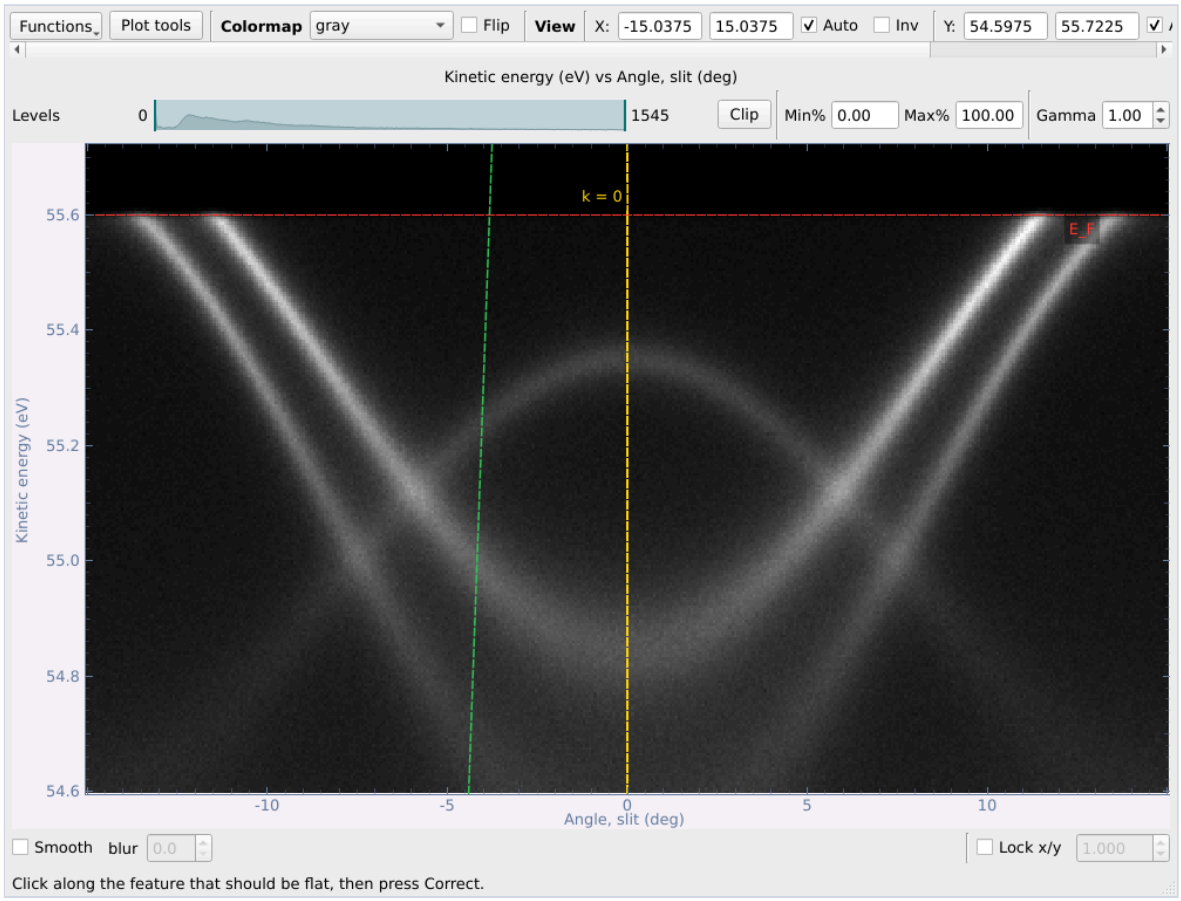
Viewer	Functions
map / kz map / k-map (contour)	Arbitrary cut; Slice: save / freeze the slice
cut	Straighten bands (FS correction); Slice: save / freeze
a map's slit cut	Straighten bands (FS correction) for the whole cube; Show the map
curve	Curve fit, Spin analysis; Crop, Bin, Normalise, Background, Shift axis, Add √N errors

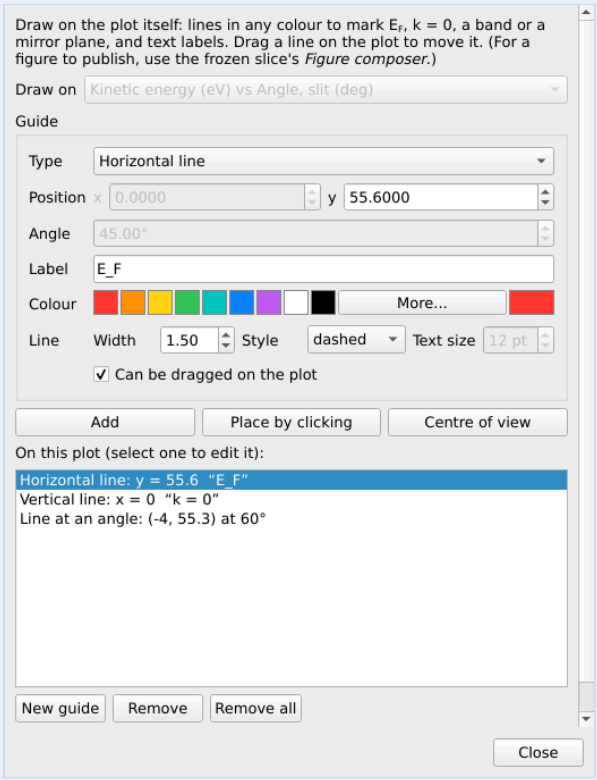
An entry that applies but cannot run yet is greyed, with the reason in its tooltip.

Brillouin zone tick box. A k-map or converted kz map on which a zone has been plotted (§8.2) keeps it: its viewer has a **Brillouin zone** tick box on the top row that shows or hides it.

Plot tools: guide lines and labels on the plot

Plot tools on the top row of every viewer (cuts, maps, spatial scans, curves, frozen slices) opens a small window that draws **on the plot in front of you**:





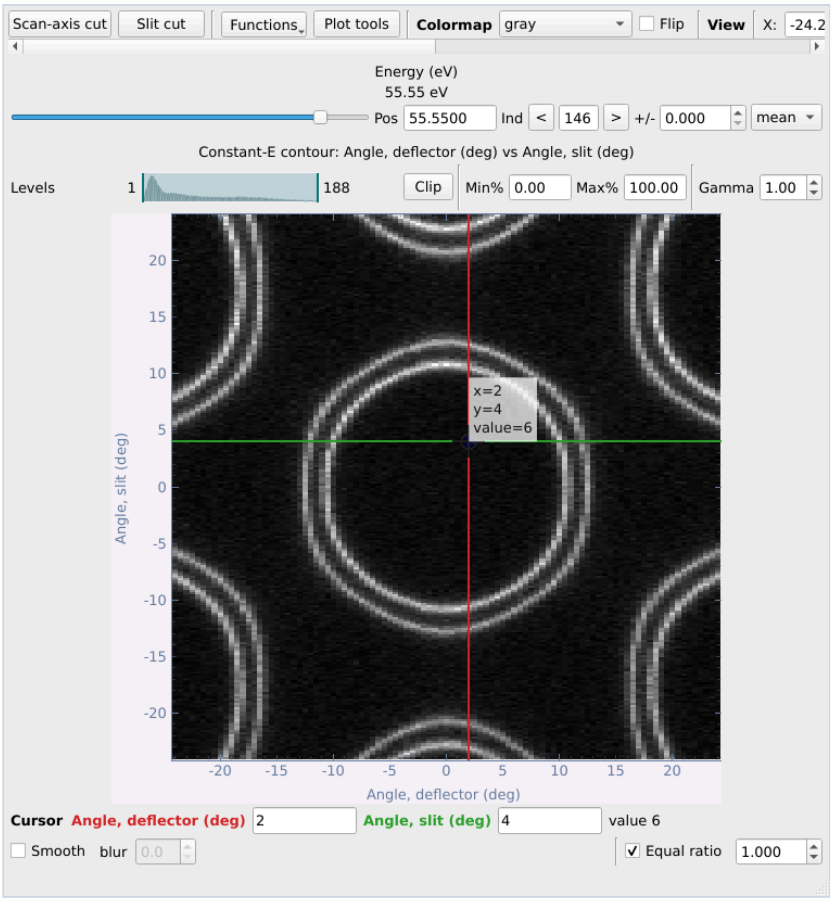
- **Type**: *Vertical line*, *Horizontal line*, *Line at an angle* (through a point; the angle is in the plot's own units, i.e. the slope is tan(angle) y-units per x-unit), or *Text label*.
- **Position** (x, y), **Angle**, an optional **Label** (shown on the line, e.g. `E_F`, `Γ`), a **Colour** (nine one-click colours, or **More...**), **Width**, **Style** (solid, dashed, dotted, dash-dot) and **Text size**.
- **Add** draws it at the position typed (**Centre of view** fills that in); **Place by clicking** draws one wherever you click on the plot.
- **Drag** a line on the plot to move it; the window follows. Select a guide in the list to **edit** it — every change shows at once — or **Remove** it; **New guide** stops editing; **Remove all** clears the plot.
- With two panels (a spatial scan), **Draw on** chooses which.
- Guides stay while the slice moves (an E_F line stays at E_F as you scroll through a map) and while the Plot tools window is closed. The panel's **Export → Image** draws them too (*Include the Plot tools guides*).

Plot tools are for looking at the data. To make a figure for a paper, use the **figure composer** and its **Figure tools** (§17).

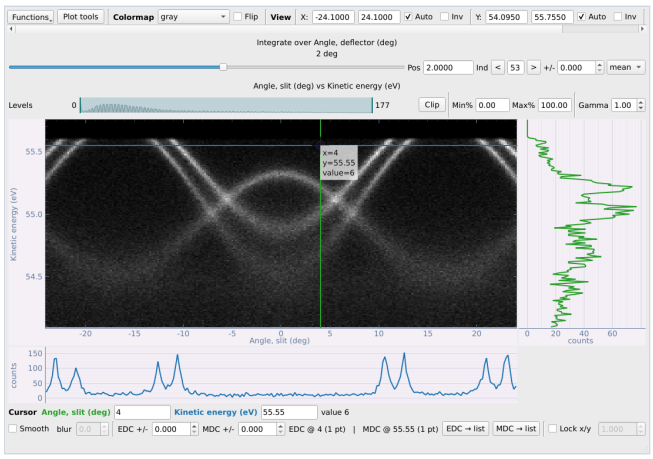
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6. Maps: the contour and its two cuts

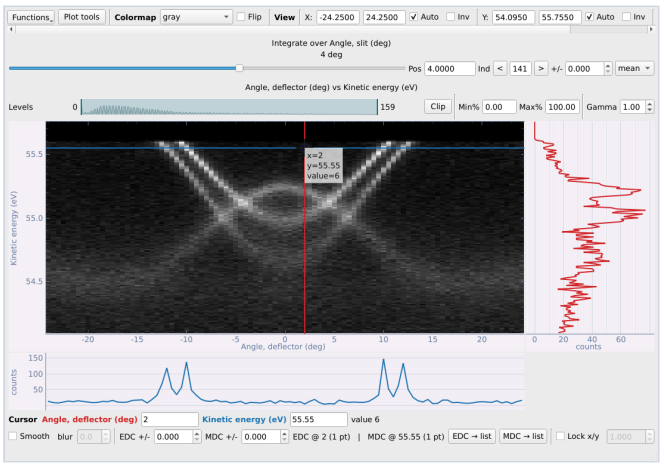
A map opens on its **constant-energy contour**. The slider at the top picks the energy (**Pos** in eV, or **Ind** as an index; **±** integrates over a window, **mean** or **sum**). **Scan-axis cut** and **Slit cut** open the two orthogonal cuts, each in its own window. (The scan axis is whatever was scanned to make the map — usually the deflector angle, but a manipulator rotation or, for a kz map, the photon energy — so the button is named for that.)



Slit cut (slit angle vs energy)



Scan-axis cut (deflector angle vs energy)



One cursor for all three windows. Switching the readout cursor on in any of them switches it on in the others, and moving it anywhere re-slices the others through the new point. The axes keep one colour each everywhere: **deflector red, slit green, energy blue**. A cursor line has the colour of the axis it holds constant, and the EDC or MDC it feeds is drawn in the same colour. Integration widths (the ± boxes) belong to each window and are never shared.

From the list you can also open a map's cut straight away (right-click → **Open slit cut** / **Open scan-axis cut**); the contour then stays hidden until you ask for it (*Functions* → *Show the map*).

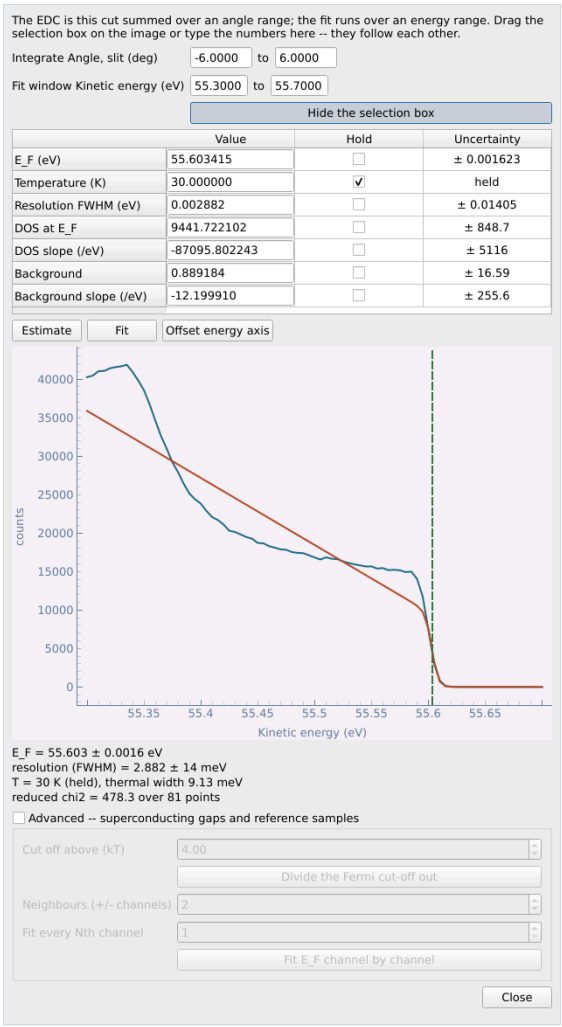
7. Cuts: from raw spectrum to band parameters

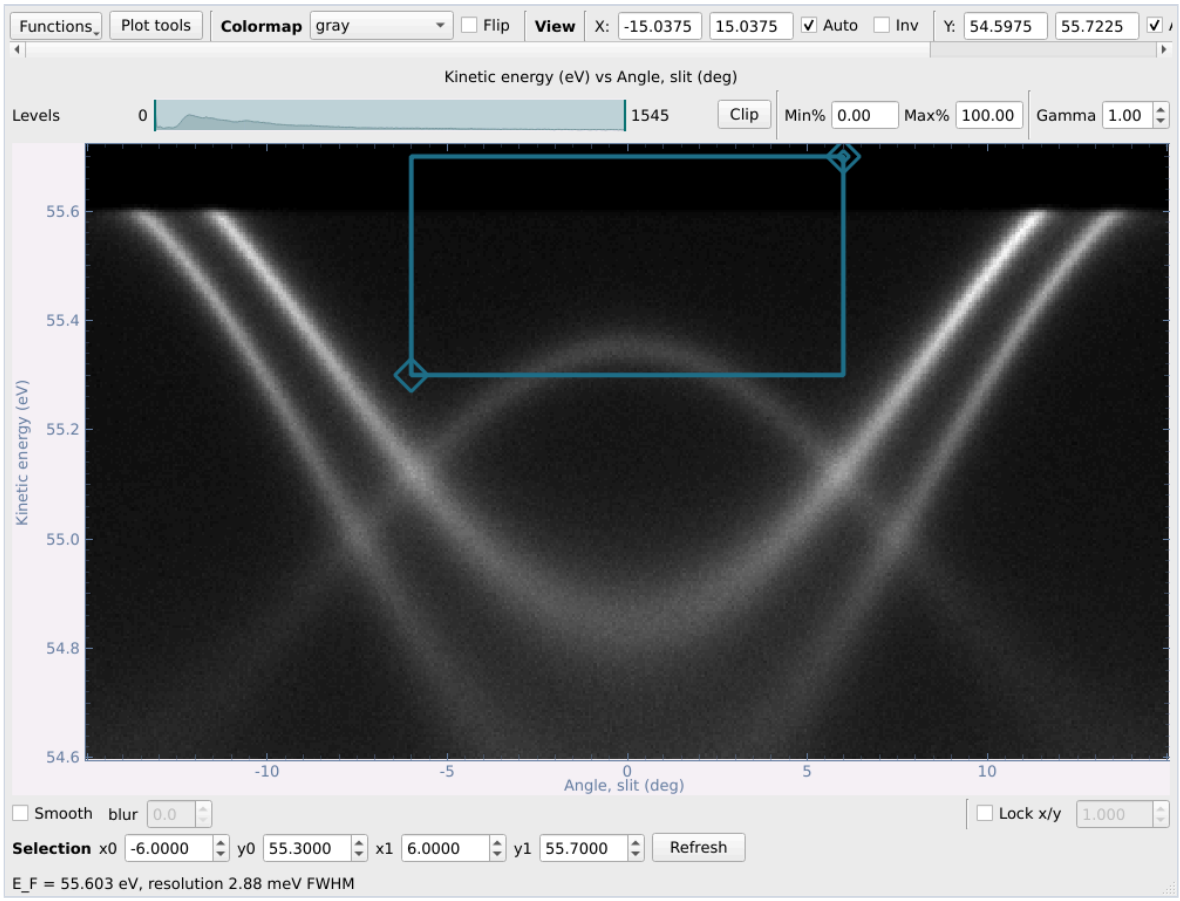
The usual order for a single cut is: **Fermi edge fit → offset the energy axis → convert to momentum → MDC/EDC fit**. Each step is a button of the tool panel: select the cut in the list and press it.

7.1 Fermi edge fit

Tool panel → Basic → **Fermi edge fit...** fits a Fermi edge (Fermi–Dirac × linear DOS, convolved with a Gaussian resolution) to the EDC summed over a box. It opens the cut's viewer together with the fit window; the two belong together, and closing either closes the other.

1. Press **Show the selection box** and drag the box on the cut over the edge (or type the angle and energy ranges). The angle range says which channels are summed; the energy range is the fit window.



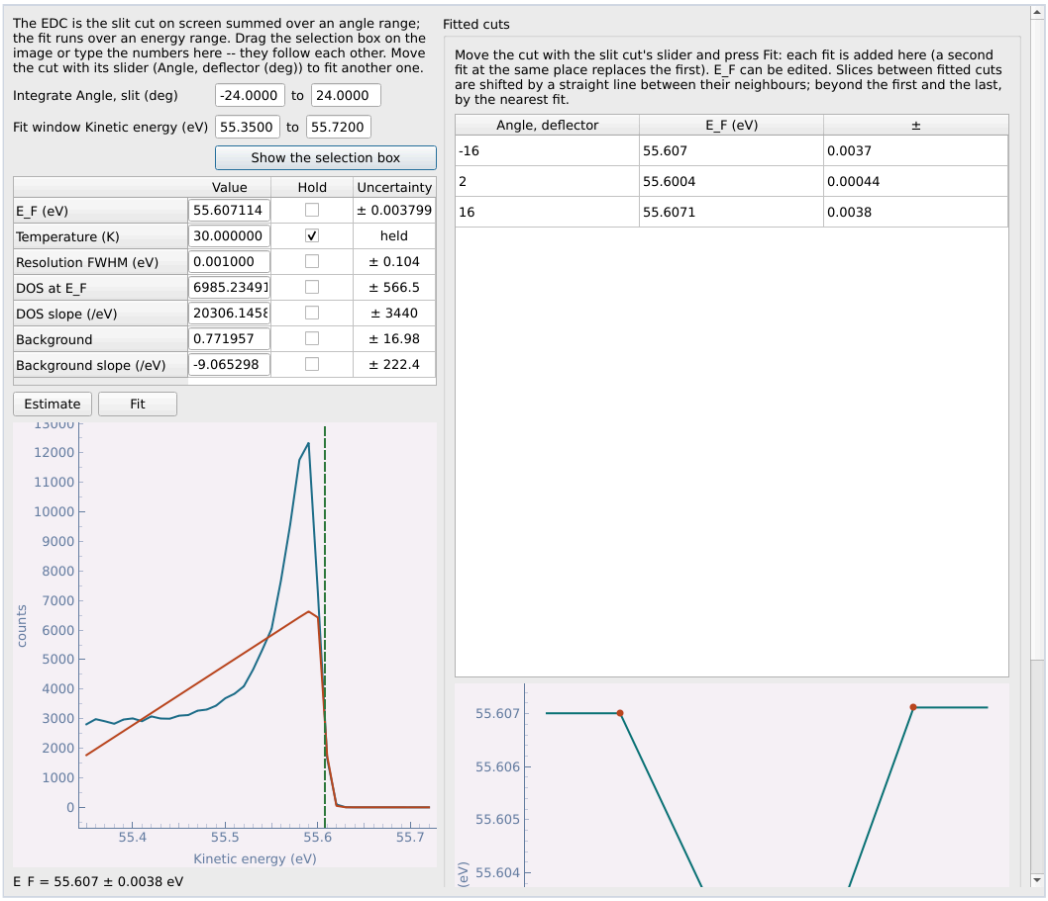


2. **Estimate**, then **Fit**. Tick **Hold** to keep a parameter fixed (the temperature is held at the recorded value by default). Every parameter is reported with its uncertainty.

3. **Offset energy axis** lists a copy of the cut on **E − E_F**.

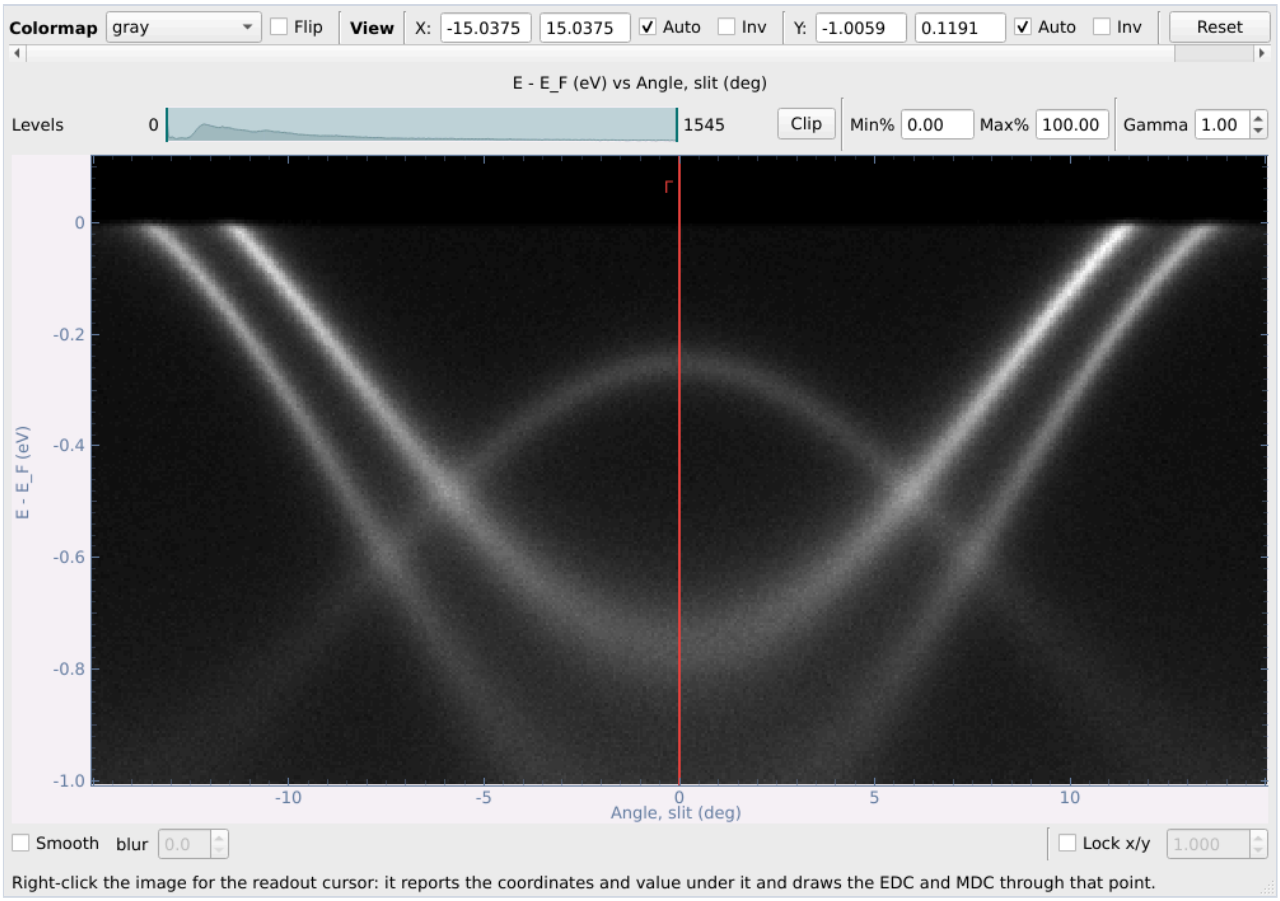
The *Advanced* corner divides the Fermi cut-off out (for gaps and reference samples) or fits E_F channel by channel.

On a map. Select a map (angle map or k-map) and press the same button: the map's **slit cut** opens with the fit window, and the slit cut's slider picks which cut is fitted. After a fit, **Offset energy axis** asks whether to shift **the whole map by this cut's E_F** (*Yes*) or to fit **one E_F per cut** (*No*). With one per cut, the **Fitted cuts** list on the right fills as you go: move the slider, press **Fit**, repeat. A second fit at the same place replaces the first, and every E_F in the list can be edited by hand. Cuts between two fitted ones are shifted by a straight line between their neighbours (beyond the first and the last, by the nearest fit), shown in the plot under the list. **Offset energy axis by these fits** lists the corrected map.

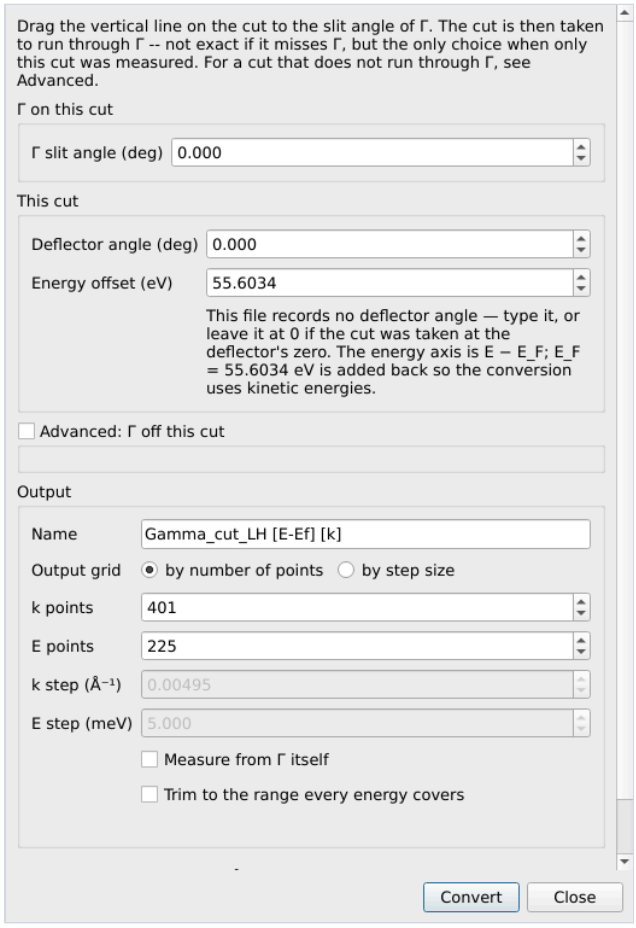


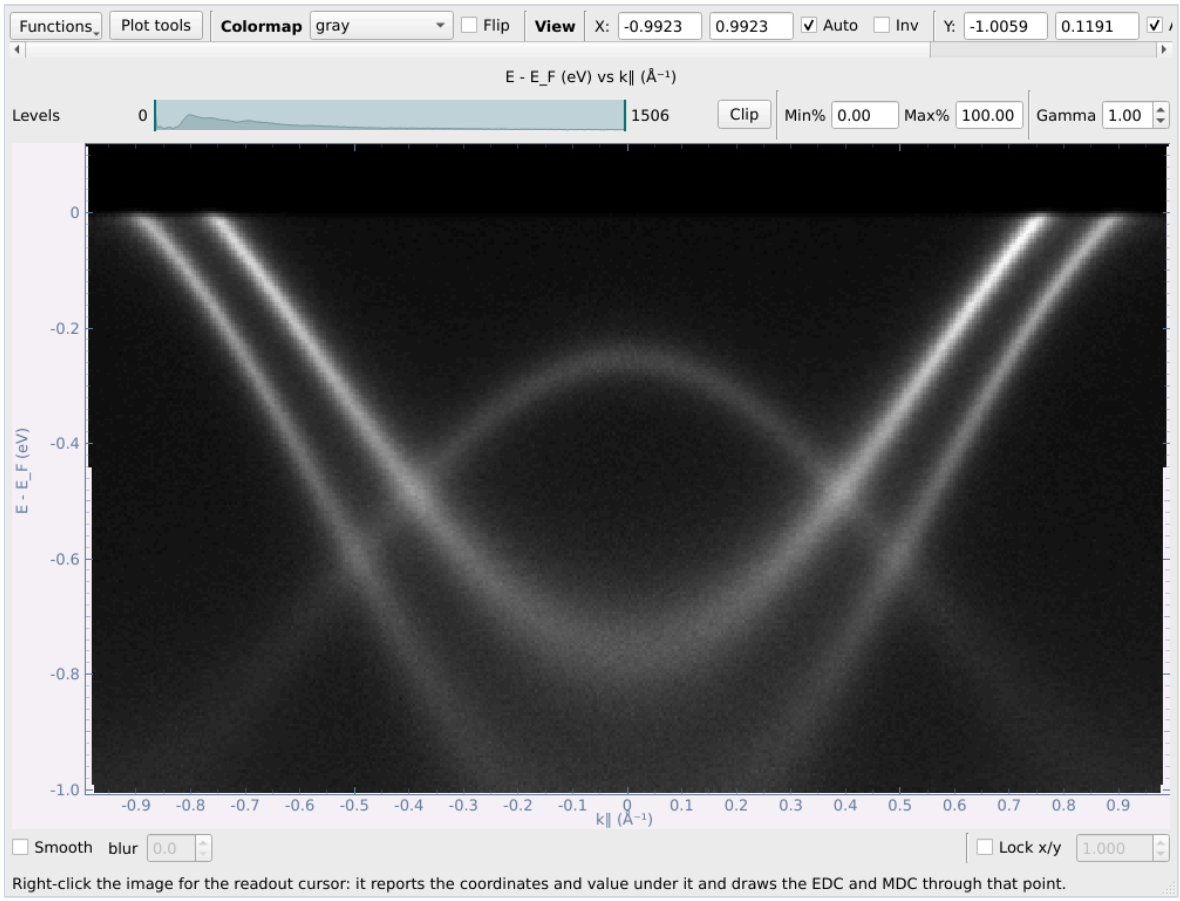
7.2 Converting a cut to momentum

Tool panel → Basic → **k conversion (cut)...** converts the angle axis to k∥ (Å^{-1}). A **vertical line** is drawn on the cut: drag it to the slit angle of Γ. The cut is then taken to run through Γ — not exact if it misses Γ, but the only choice when only this cut was measured.



- **Advanced: Γ off this cut** is for a cut that does not run through Γ: type the deflector and slit angles of Γ, or **From a converted map…** to inherit them from a map of the same sample already converted.
- If the cut is already on E − E_F, **Energy offset** is filled in with the E_F that was subtracted (the note under it says so): the conversion uses true kinetic energies, and the result stays on E − E_F.
- **Remembered settings.** The dialog starts from the last conversion of a cut of the same size (same number of points on each axis) — the usual case for a series of cuts of one sample. The map and kz conversions do the same.

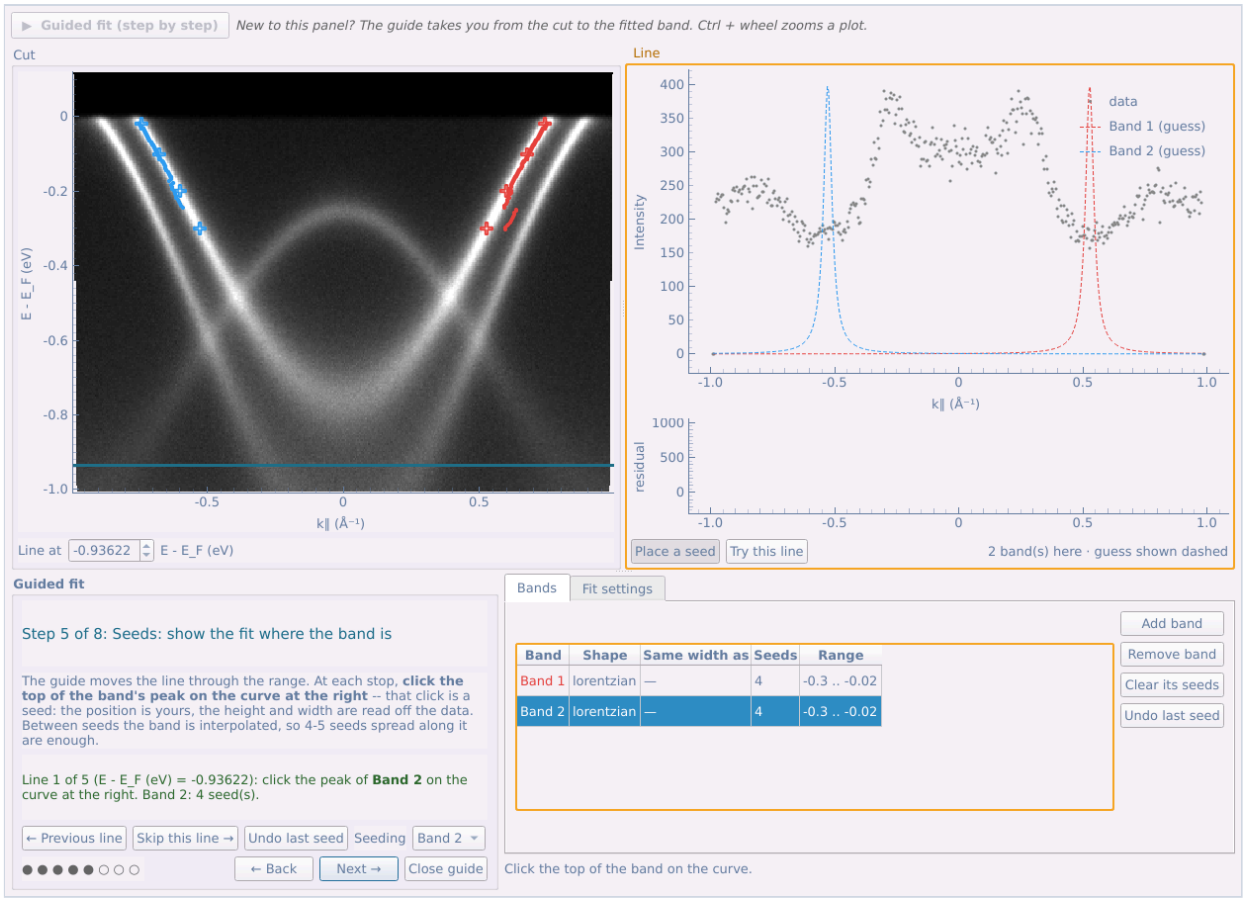




7.3 MDC fit and EDC fit

Two tools of the tool panel's Advanced tab fit peaks line by line:

- **MDC fit (k-cut only)...** fits the momentum distribution curves and reads the band off the fitted centres: Fermi velocity, effective mass, self-energy. It only takes a cut whose axis is already in $Å^{-1}$ — a velocity in eV·deg would be meaningless — and its button is greyed otherwise.
- **EDC fit...** fits the energy distribution curves, on any cut, optionally with the Fermi cut-off in the model. v_F and m* from the fitted band need a cut in momentum.

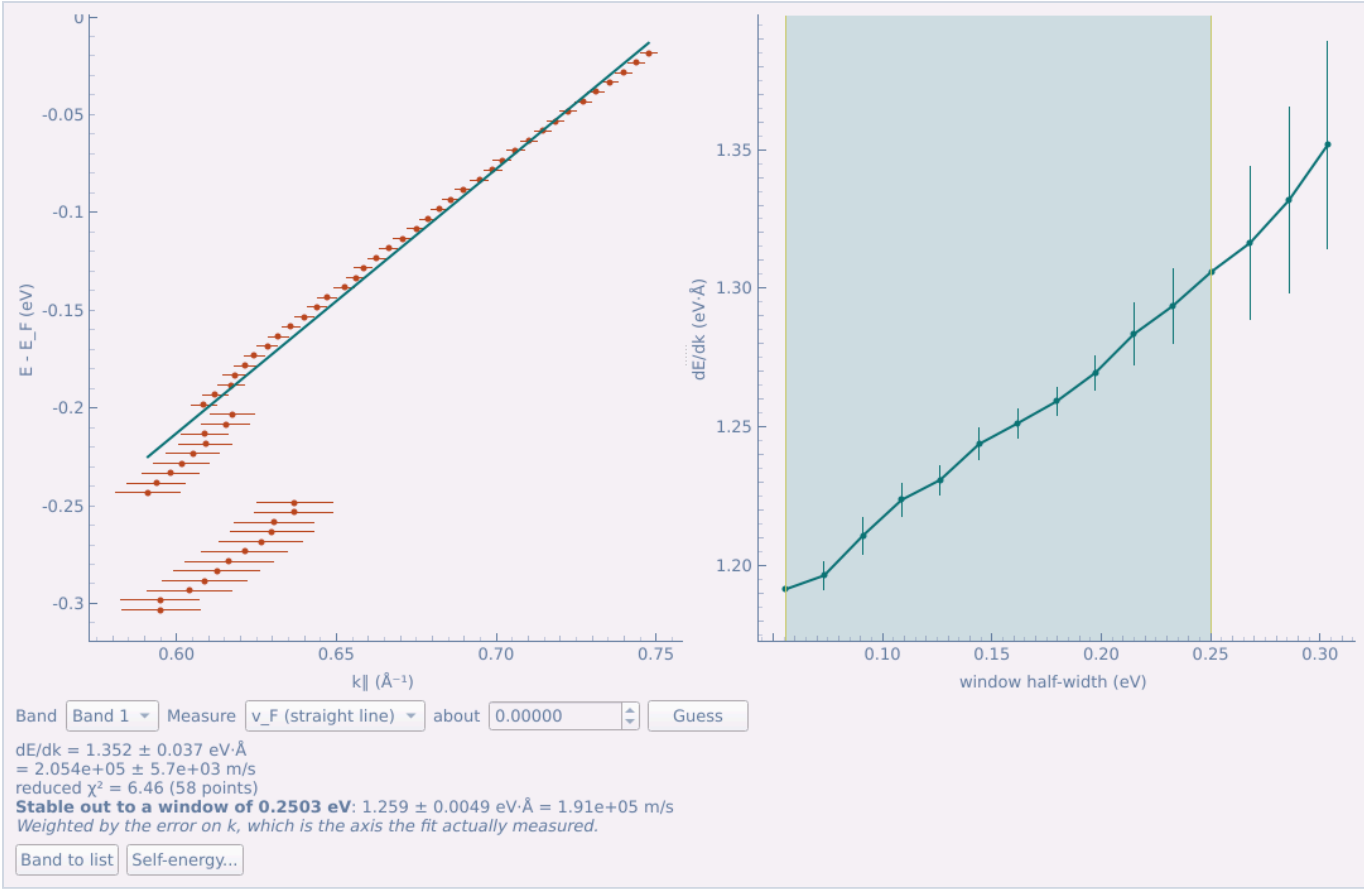


Guided fit (step by step), at the top of the panel, walks through the whole fit beside the controls, one step at a time, saying what each step is for and outlining the part of the panel it uses: what an MDC (EDC) fit does, the line shape, the range (*Start here* / *End here* at the blue line), the bands, the seeds (the guide moves the line to five places in the range — click the band's peak on the curve at each), one test line, the series, and the result. *Next* waits until a step has what the fit needs.

The pictures and the controls share the window with a handle between them, so the settings are always in sight; the mouse wheel over a picture scrolls the window when it is too short for everything (Ctrl + wheel zooms the picture).

By hand:

1. **Bands** tab: **Add band** for each branch; select a band, press **Place a seed** and click the peak on the curve (right). Four or five seeds down a band are enough; **Line at** moves the line being looked at.
2. **Try this line** shows the fit of the current line (dashed: the guess).
3. **Fit settings** tab: peak shape (Lorentzian, Gaussian, Voigt), background, step, then **Fit the series**. Fitted positions are drawn on the image, sized by amplitude.
4. **Dispersion...** fits the band: **v_F** (straight line), **m*** (parabola) — and checks how far out the fit is stable. **Self-energy...** gives Re Σ and Im Σ.

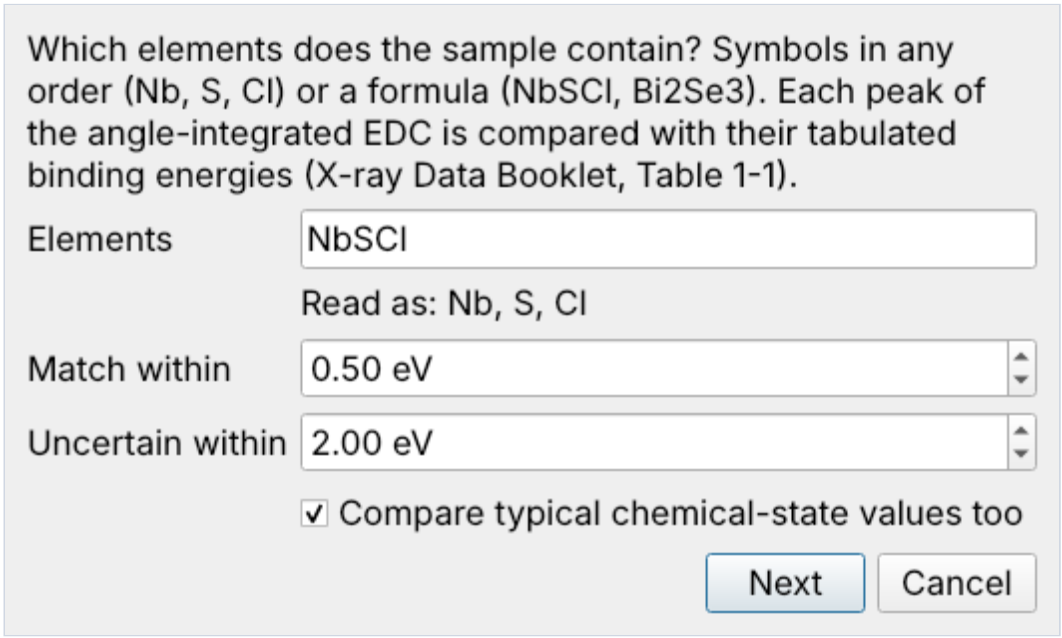


Save table... writes every fitted line; **Band to list** puts a band in the main list.

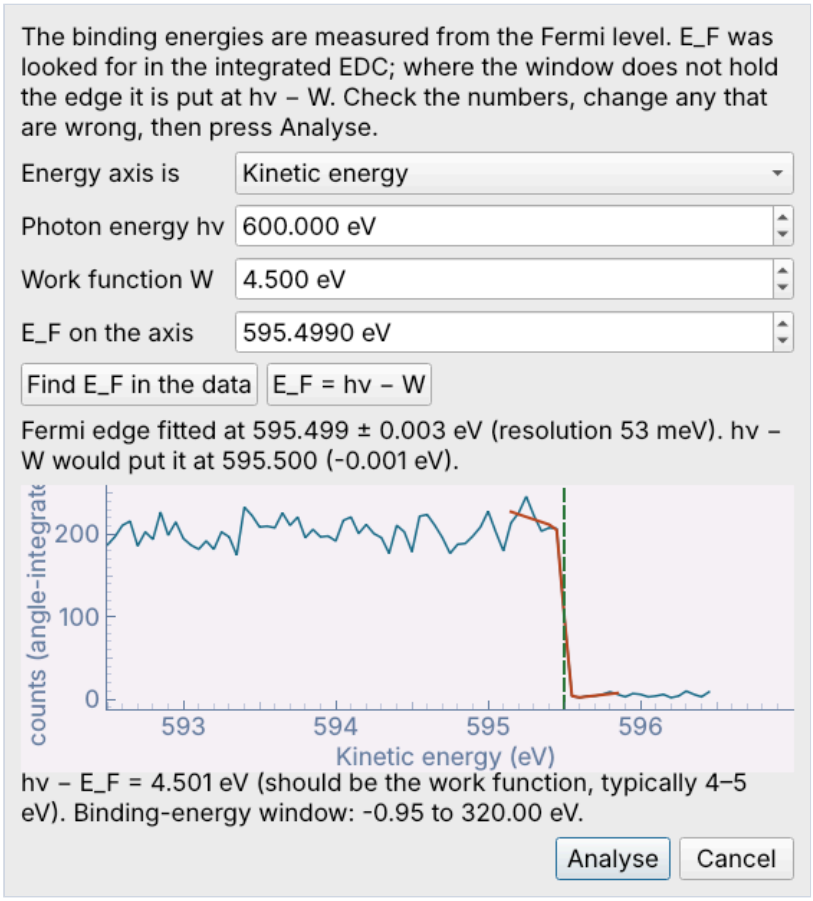
7.4 Core-level check

A wide scan of the core levels is listed as a cut too. Select it and press tool panel → Basic → **Core-level check...** to find out what its peaks are.

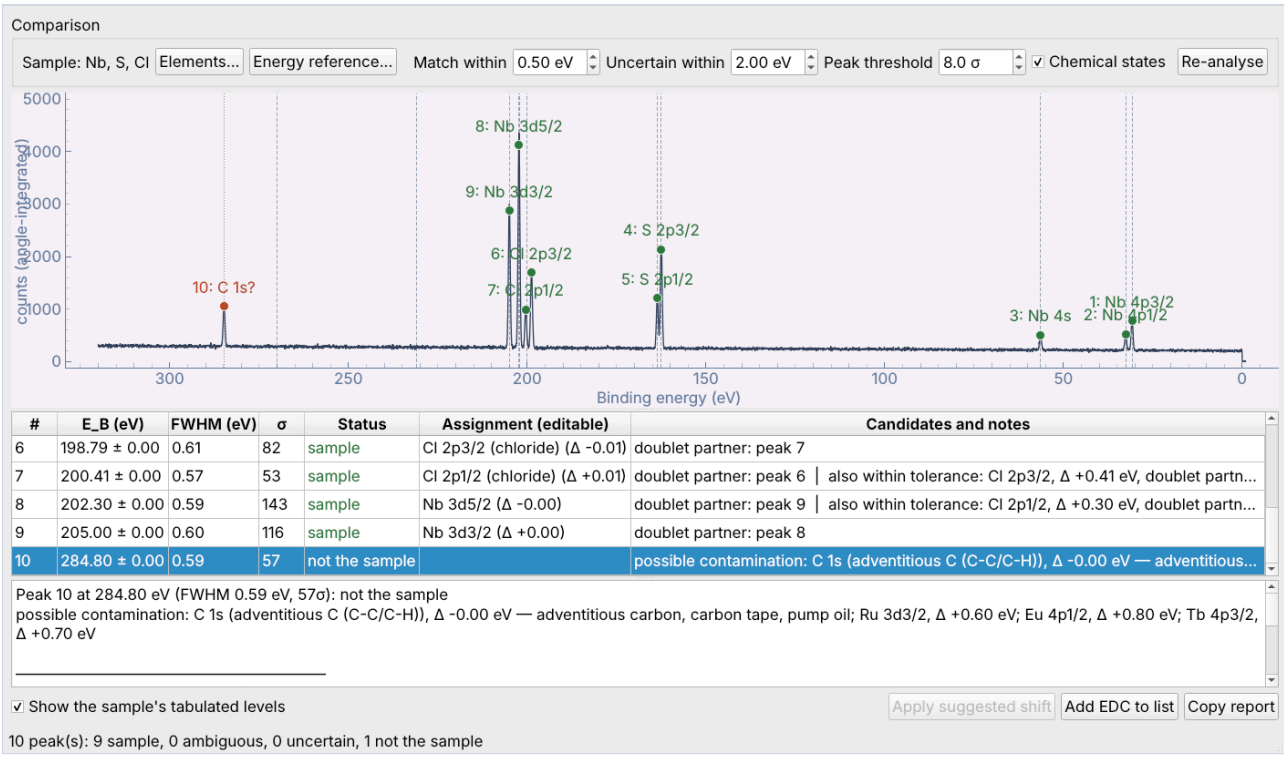
1. Type the sample's elements — `NbSCl` or `Nb, S, Cl` — and press **Next**. *Match within* (0.5 eV) is how close a peak must be to a tabulated level to count as the sample's.



2. Check the energy reference. E_F is fitted when the window holds the Fermi edge, otherwise put at hν − W. Change the photon energy, the work function or E_F if they are wrong (the green line can be dragged); a hν − E_F far from 4–5 eV means the photon energy is off. Press **Analyse**.



3. The results: the EDC on a binding-energy axis, each peak marked — green *sample* (orbital named), orange *uncertain* (a level of the sample within 2 eV: chemical shift or charging — your call), red *not the sample* (common elements that could explain it: contamination, higher-order light, Auger lines). Click a row for the full list of candidates. Type your own assignment in the *Assignment* column.



Weak lines and peaks by hand. Tick **Log counts** to see weak lines next to a strong one; **Min height** (default 0.2 % of the strongest) sets how small a found peak may be. To add a peak the search missed, press **Click to add peak** and click the plot (a double-click always works), or type its binding energy after *or at* and press **Add**: its position is refined by a local fit, it is marked ◆, and it is named with every tabulated level within ±2 eV. **Remove selected** drops peaks; **Reset peaks** brings back the found ones and forgets those added by hand. Assignments you typed survive a **Re-analyse**.

Peak table... lists every tabulated level, filtered by element and binding-energy range: look a peak up, show a level on the plot, or add it as a peak with that level as its assignment. **Select range** shows a band you drag over a region; **Zoom window...** opens it (or the current view) in a window of its own with a cursor, a Gaussian fit at the cursor, and the negative second derivative divided by the intensity — which shows a shoulder or a weak line beside a strong one as clearly as the strong one.

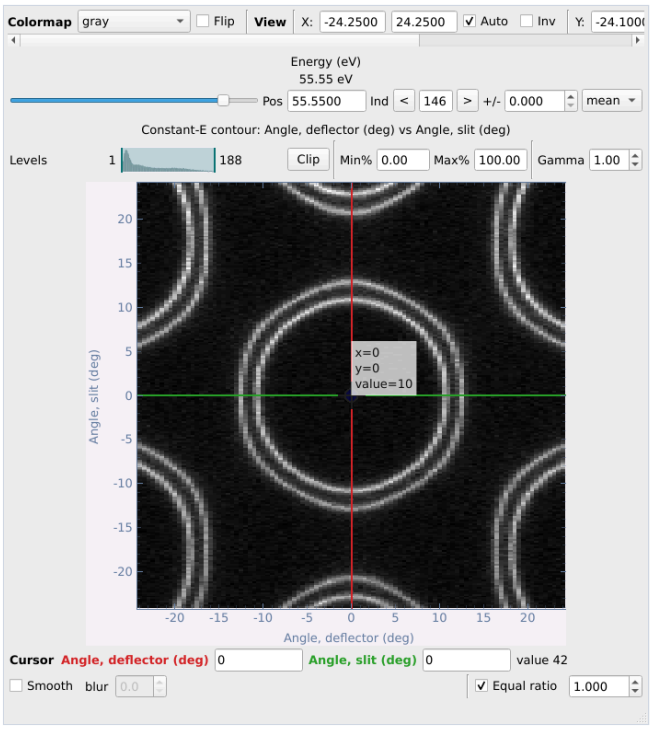
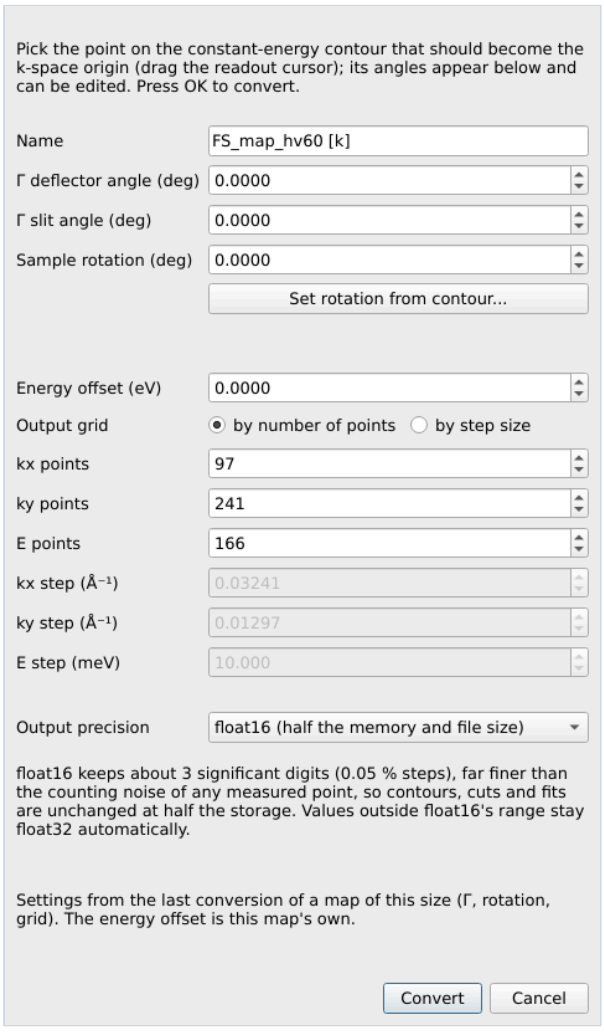
If every peak is off by the same amount, the notes say so and **Apply suggested shift** corrects it (a charged sample, or a wrong E_F). **Add EDC to list** lists the EDC on the binding-energy scale; **Copy report** copies the whole result as text. Measuring at a second photon energy settles what stays doubtful: core levels stay put in binding energy, Auger lines and higher-order light move.

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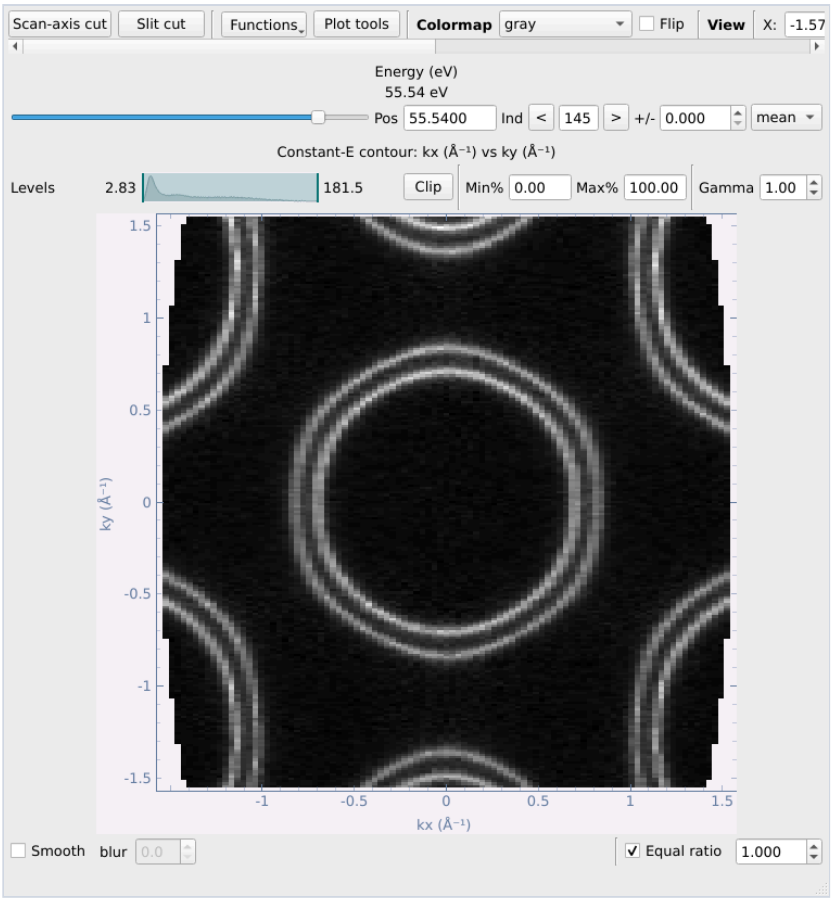
8. From angles to momentum for a map, and the Brillouin zone

8.1 Converting a map to momentum

Select an angle map and press tool panel → Basic → **k conversion (map)...**. The map opens on a bare constant-energy contour — no cuts, no other functions, just the picture — next to the conversion window; the two close together. Drag the readout cursor to the point that should become **k = (0, 0)** — the dialog follows it — then **Convert**. The result is a new `k-map` row in the list.

The contour for picking Γ	The conversion
	

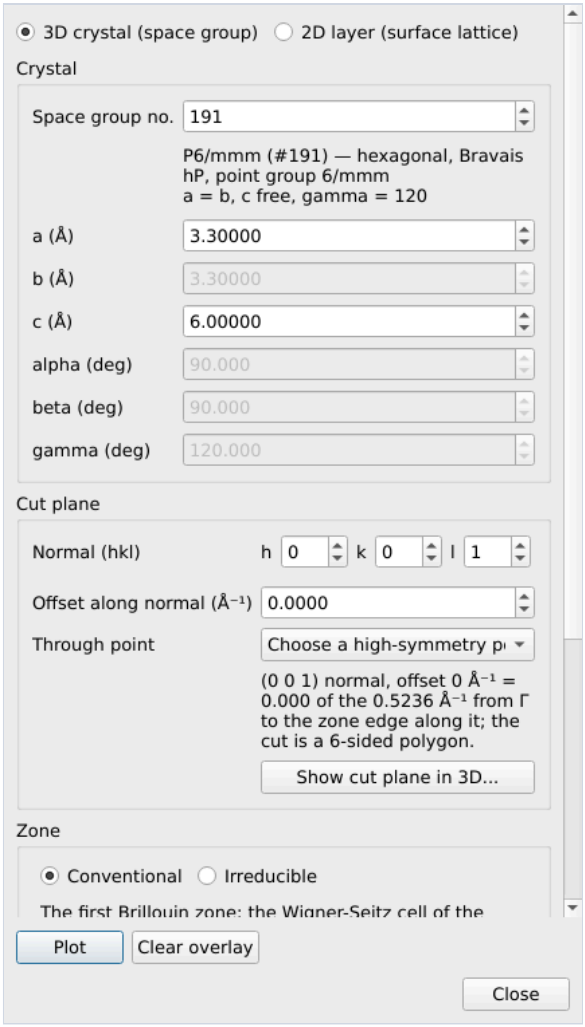
- **Γ deflector angle / Γ slit angle**: the angles of the point that becomes k = (0, 0), filled in from the cursor.
- **Sample rotation** turns the map about the surface normal; **Set rotation from contour...** reads it off a direction you click on the contour (two points along a mirror line, say).
- **Output grid**: number of points, or resolution (Å⁻¹, meV). The numbers of points start at the measured ones — as many kx points as deflector angles, ky points as slit angles, E points as energies.
- **Energy offset** is added to the energy axis before converting, for an axis that is not a true kinetic energy (filled in automatically for a map on E − E_F).
- The settings start from the last conversion of a map of the same size, so a second map of a series needs no retyping.



8.2 Brillouin zone

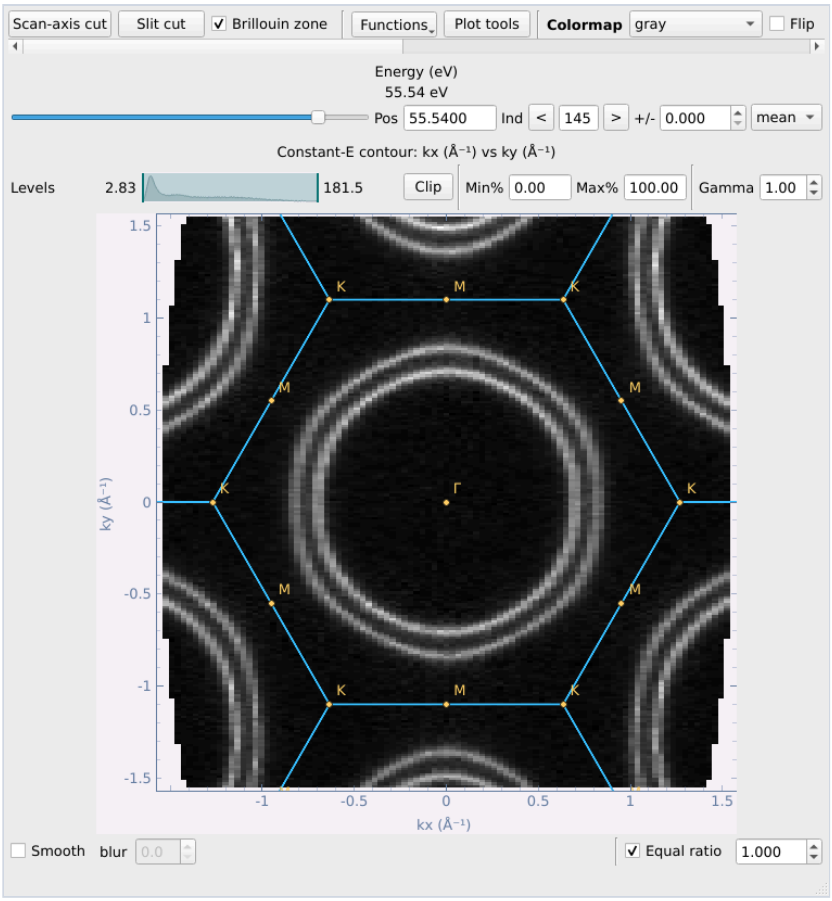
Select a k-map (or a converted kz map) and press tool panel → Basic → **Brillouin zone...**. Like the map conversion, it opens a contour of its own with the zone window, and the two close together.

1. **Space group** first (1–230); the lattice parameters it fixes are greyed and follow the free ones. Or switch to **2D layer** for a surface lattice.
2. **Cut plane**: the surface normal (hkl) and an offset along it — the cut of the 3-D zone by that plane is drawn, for any surface, not only (001). **Through point** lists the offsets at which the plane passes through a high-symmetry point (Γ, Z, L, F ... for a rhombohedral crystal); choose one rather than working the number out. **Show cut plane in 3D...** previews it, with the named points marked and the ones in the plane ringed; **Label high-symmetry points** writes them on the contour, which is the quickest way to set the azimuth. The dialog opens on the last crystal you used, so a second map of the same sample needs no retyping.
3. **Conventional** or **Irreducible** zone; **Azimuth** (or **Pick direction from contour...**); **Repeat across the field of view**.
4. **Moiré** adds a second, twisted or mismatched layer and its moiré zone.
5. **Plot**.

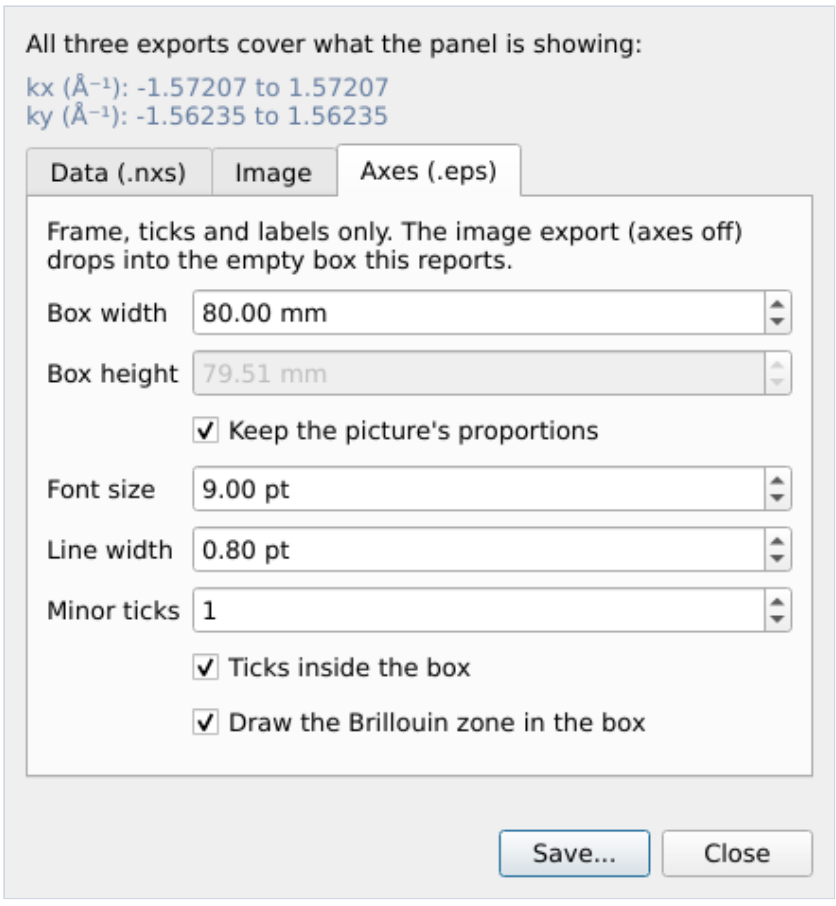


The window is tall; it scrolls, and can be resized like every other window.

The zone stays with the dataset. Once plotted, it is a property of the k-map: it is saved with it into a `.nxs` file, and its viewer shows it with the **Brillouin zone** tick box on the top row. **Clear overlay** removes it.



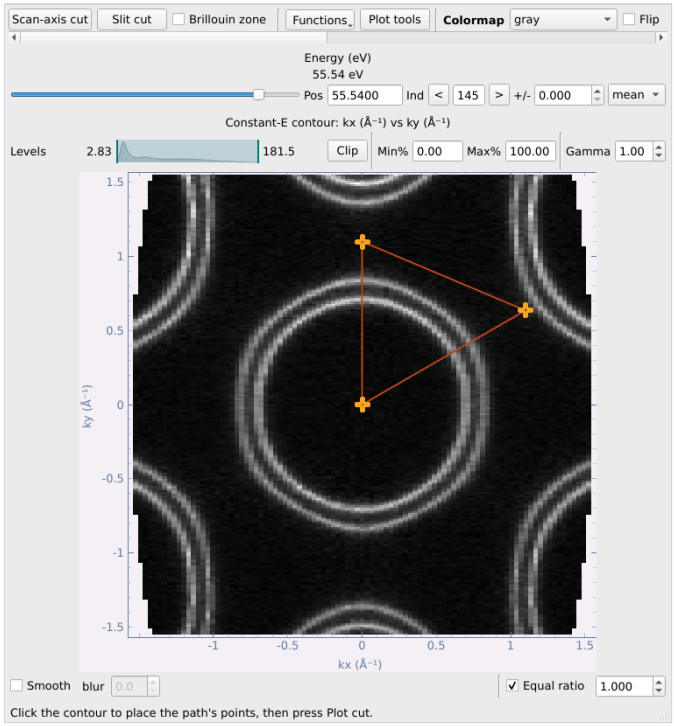
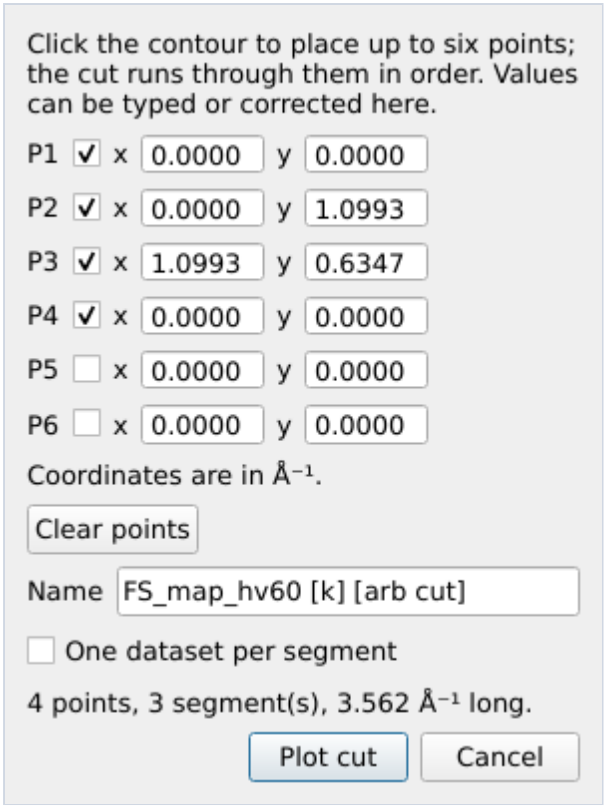
Its exports can draw it too: **Export → Image...** has **Include the Brillouin zone**, and **Axes only (.eps)...** has **Draw the Brillouin zone in the box**, so the zone comes out as vector lines over the image in Illustrator.



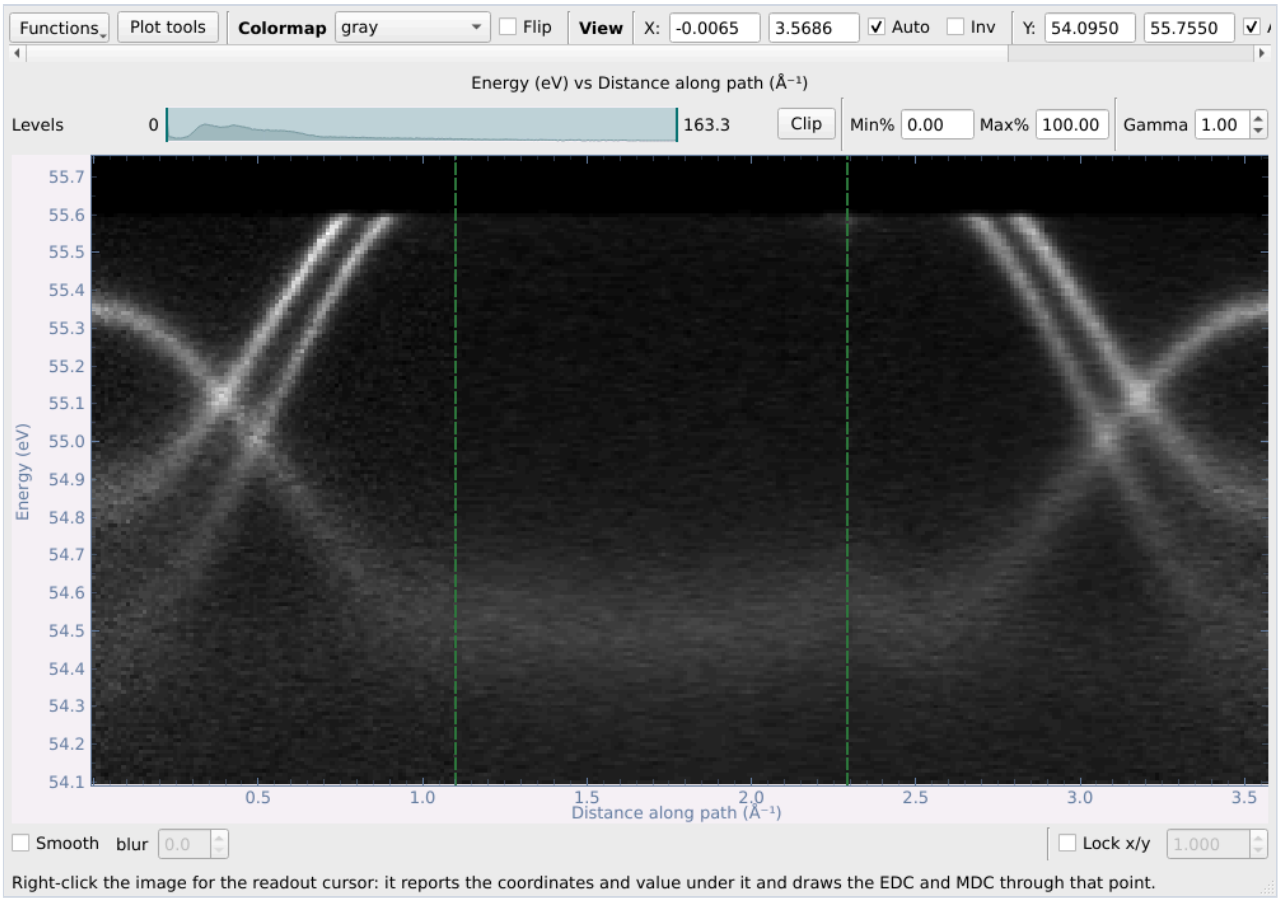
9. Arbitrary cuts and Fermi-surface correction

9.1 Arbitrary cut

Functions → Arbitrary cut… on any contour: click up to six points (or type them) and the cut runs through them in order — here Γ → M → K → Γ on the k-map. **One dataset per segment** lists each leg separately.

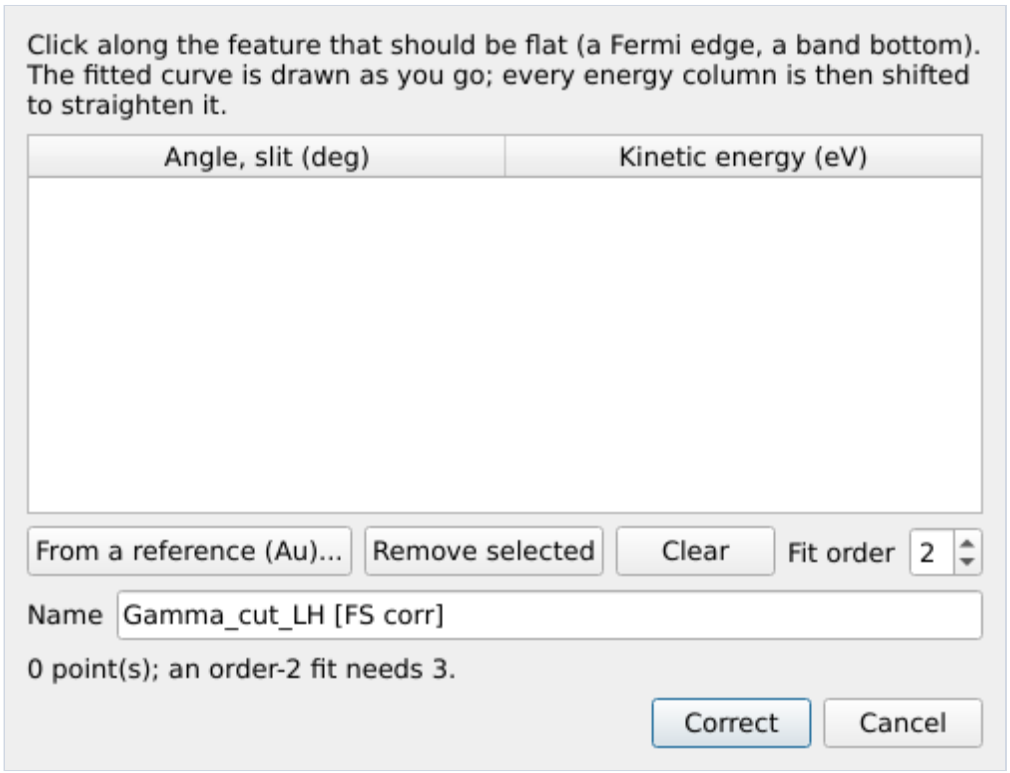
Path on the contour	Points
	

The result is a cut against the distance along the path; the corners of the path are marked with dashed green lines, so a kink there is not mistaken for physics.



9.2 Fermi-surface (FS) correction

Functions → Straighten bands (FS correction)... on a cut (or a map's slit cut) straightens a feature that should be flat — usually the Fermi edge, which curves across the analyser angle. Click along the feature; a polynomial (order 2 by default) is fitted and every angle column is shifted in energy. **From a reference (Au)...** takes the curve from a gold spectrum in the list instead. On a map's slit cut the same shift is applied to the whole cube.

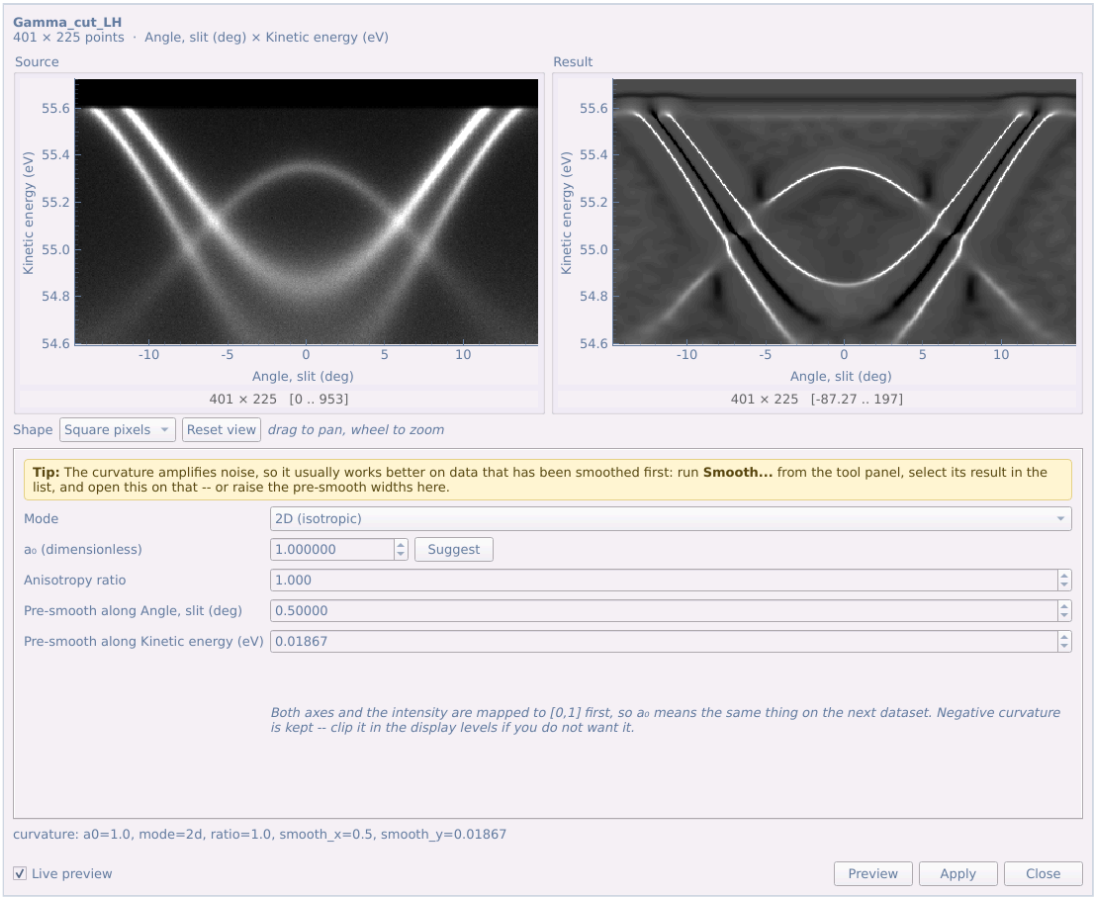


10. Processing: smoothing, derivatives, curvature

Each filter has a window of its own, opened from the tool panel with one or more cuts (or maps) selected:

Tab	Button	What it does
Basic	**Background subtraction...**	removes the intensity that is not the bands: Shirley or Tougaard inelastic background, a floor common to every angle, a smooth polynomial — or divides out the Fermi cut-off
Advanced	**Smooth...**	Gaussian, Savitzky–Golay or moving average, with optional spike removal
Advanced	**Second derivative...**	$-d^2I/dE^2$ or $-d^2I/dk^2$ after a pre-smooth, to sharpen the bands
Advanced	**Curvature...**	1-D or 2-D curvature (Zhang et al.), which follows the band better than a second derivative
Advanced	**Symmetrise...**	mirror a cut about a line; for a map, mirror or rotate its constant-energy planes

Every result is a new dataset carrying the steps that made it.

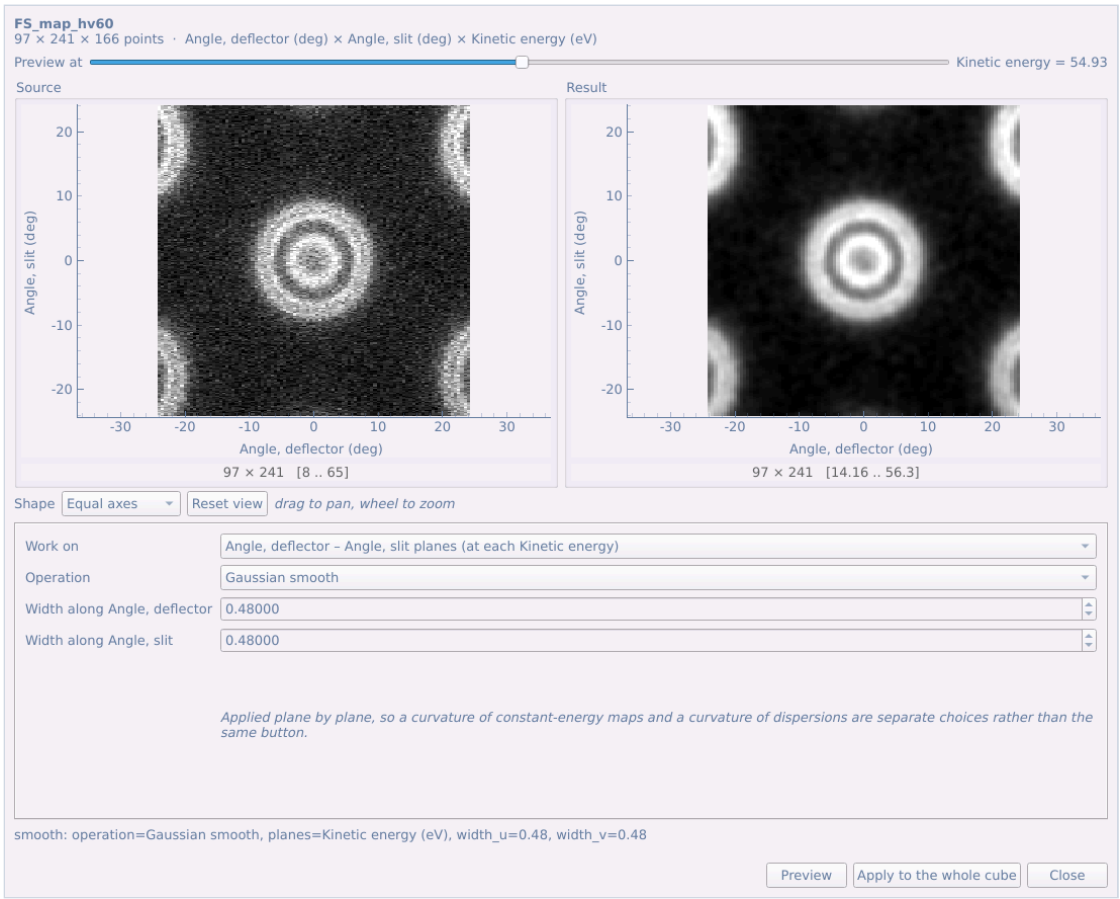


- **Source** and **Result** are shown side by side on the same scale, so you can see when a setting has eaten the band. They pan and zoom together.
- **Smooth first.** The second-derivative and curvature windows open with a tip: both amplify noise, so they usually work better on data smoothed first — run **Smooth...**, select its result in the list, and open the filter on that, or raise the pre-smooth widths in the window itself. A smoothed result

carries the tag ‹**smoothed**› in the list (and so does anything made from it); on such data the tip is replaced by a short "already smoothed" line and the pre-smooth widths start at 0.

- **Shape**: *Square pixels* (one data point is square — undistorted for an angle-vs-energy cut), *Equal axes* (one unit is the same length on both axes — right for angle-vs-angle or k-vs-k; chosen automatically when both axes have the same unit), or *Fill the box*.
- **Live preview** re-computes as you change a value; **Apply** lists the result.
- Starting values: **Symmetrise** mirrors about a vertical line through angle (or k) = 0, i.e. k → −k; the **Fermi cut-off** under *Background subtraction* starts at the E_F the file records, or else at the steepest fall of the angle-integrated EDC, with the file's temperature and resolution.

For a map the window works plane by plane, and asks **which planes** first — constant-energy maps or dispersions — since the curvature of a contour and the curvature of a cut are different quantities. **Preview at** picks the plane shown; **Apply to the whole cube** runs it on every plane. The operations and widths are named after the chosen plane's own axes (*Second derivative along Kinetic energy*…), and the smoothing widths start at 1/100 of each axis.

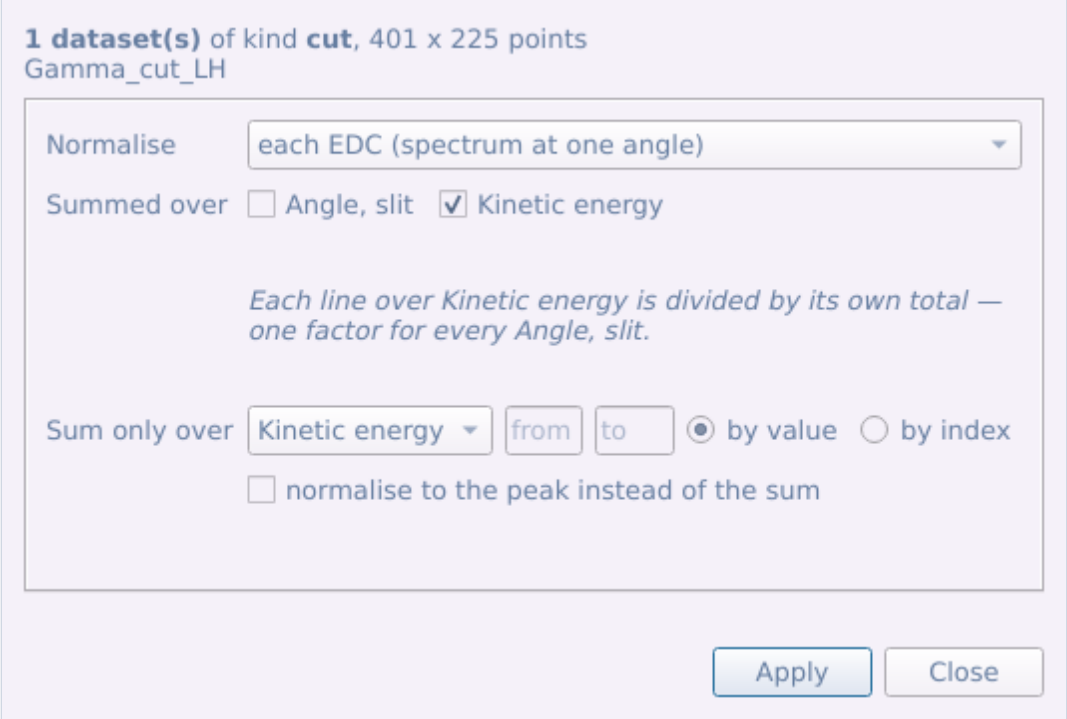


Truncate..., **Compress...** and **Self-normalise...** (under the list) change the grid or the scale of one or several datasets at once — as long as they have the same kind and the same number of points on every axis: **Truncate** crops to a range, **Compress** bins points together, **Self-normalise** divides each spectrum by its own total.

Self-normalise divides each piece of the data by its own total (or peak):

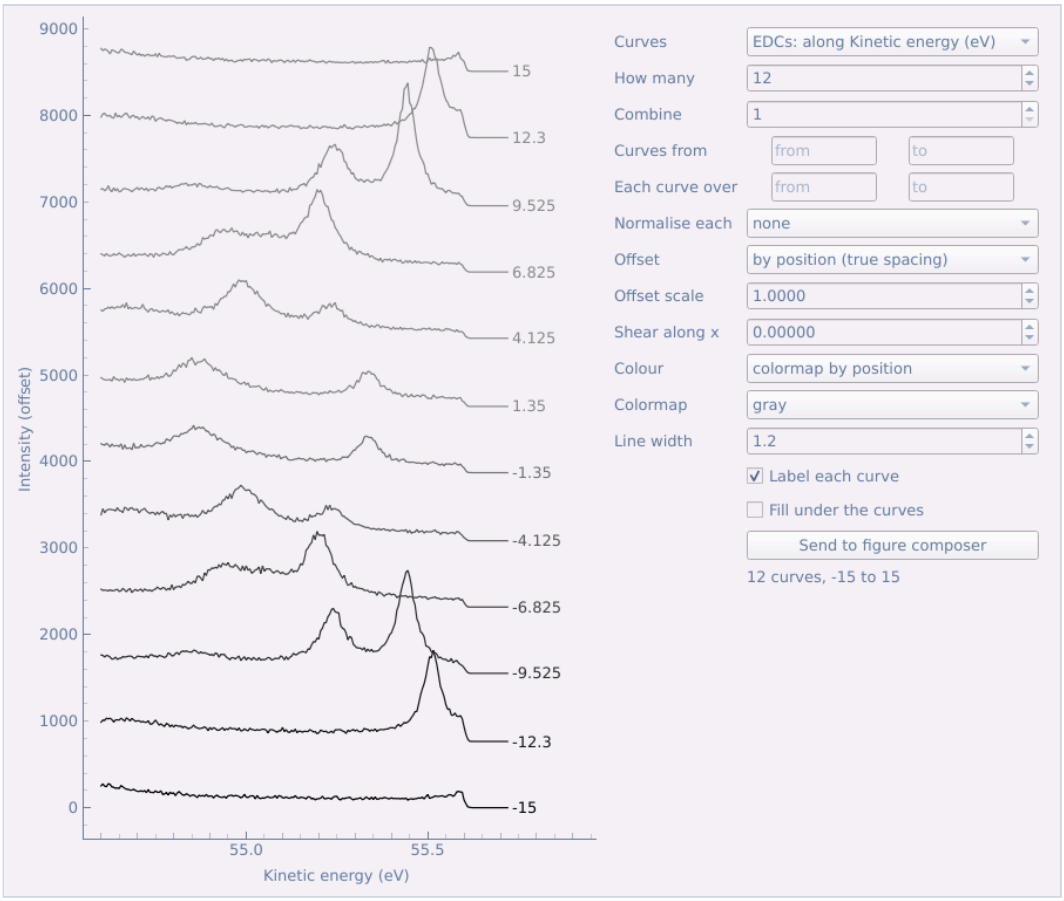
- **Normalise** picks the piece in ARPES terms, and starts on the usual one: *each EDC* for a cut, *each cut (slit × energy at one scan step)* for a map, k-map or kz map — this takes a flux or beam-current drift along the scan out. Other choices: each MDC, each EDC of a map, each constant-energy map, or *custom*.

- **Summed over** shows the axes this sums over; the ticks can be changed by hand, and the sentence under them says what the current ticks do ("Each plane over Angle, slit × Kinetic energy is divided by its own total — one factor for every Angle, deflector").
- **Sum only over** restricts the sum to a window on one of the summed axes — the energy by default — for example the energies of a core level.



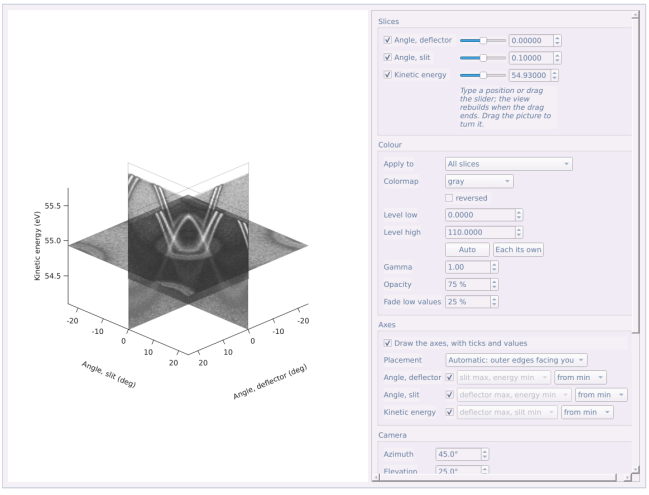
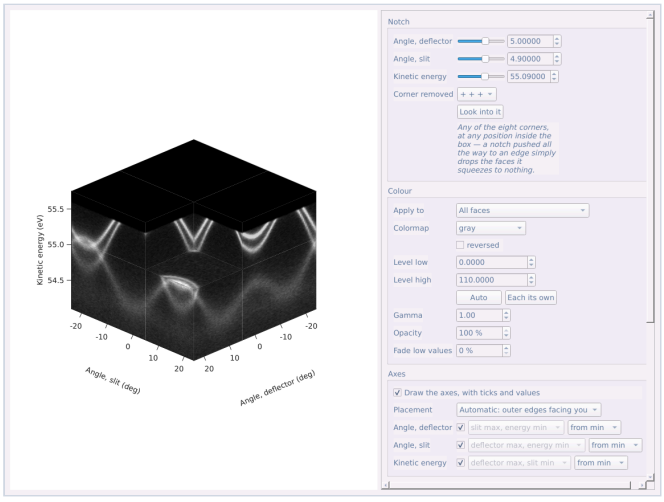
11. Stack plots and the 3-D views

Stack plot (EDC / MDC)... (tool panel → Advanced, with a cut selected): a waterfall of EDCs or MDCs, with the number of curves, binning, normalisation, offset, shear and colouring controls; **Send to figure composer** makes it a figure panel.

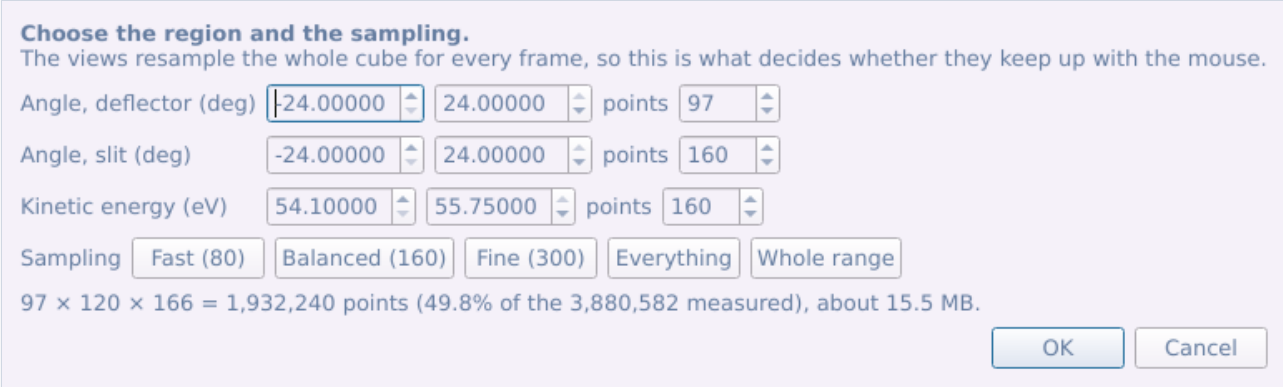


Three tools of the Advanced tab show a map in 3-D, each in a window of its own — drag the picture to turn it:

- **3D slices...** — the three orthogonal planes through the cube, each placed where you choose (tick box, slider or typed position per axis).
- **Notched cube...** — the cube with one corner cut away: a constant-energy map and two dispersions in one picture.
- **Volume projection...** — the whole volume projected along the view: maximum intensity, sum, alpha composite, or a shaded isosurface.

3D slices	Notched cube
	

- **Sampling.** Every frame resamples the cube, so each window opens at *Balanced* sampling, which keeps up with the mouse. The current sampling is written under **Save the view...**; **Change sampling...** picks the region and *Fast*, *Balanced* or *Fine*, and redraws from the full cube.

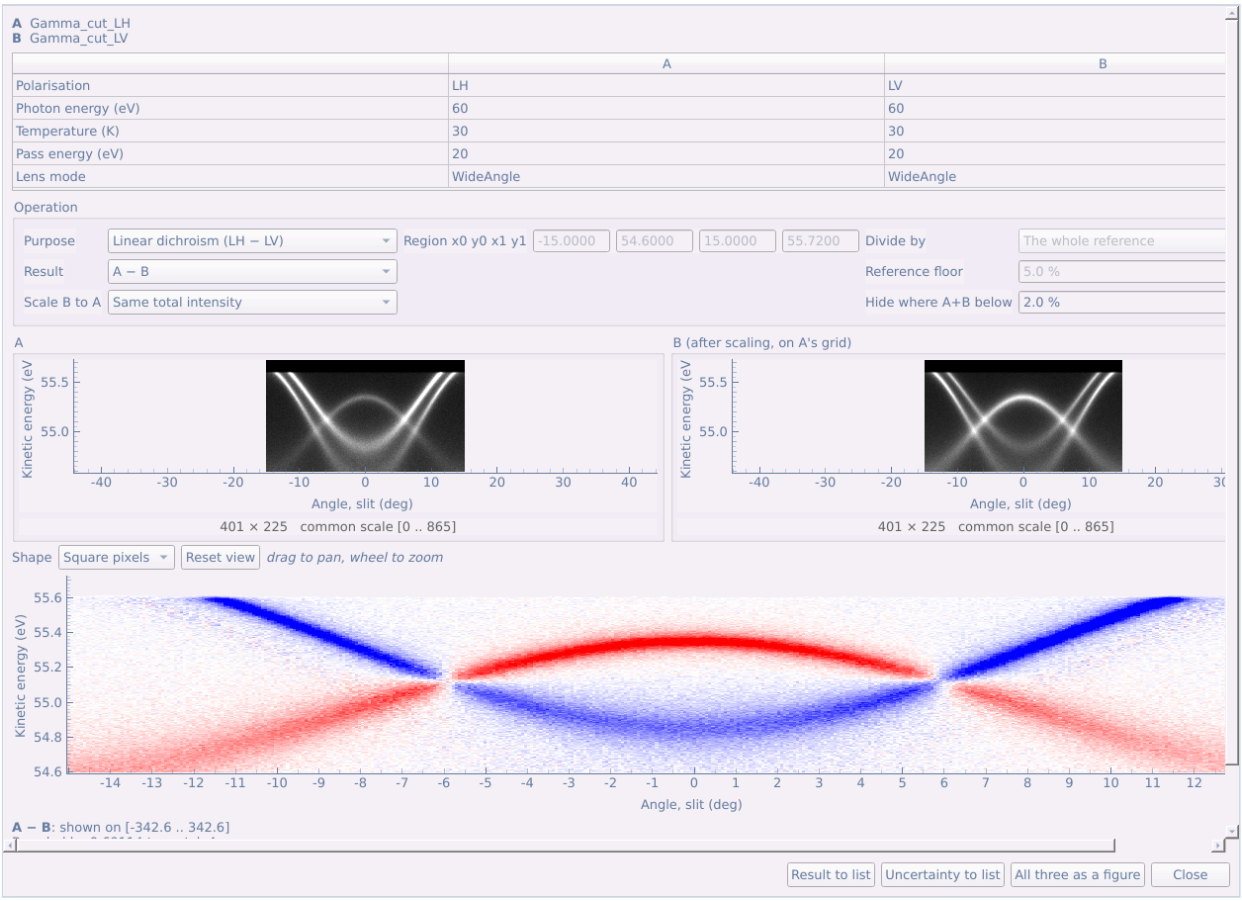


- **Colour.** *Apply to* sets the colours of **all slices** at once or of one orientation (the energy plane in one colormap, the dispersions in another): colormap (and *reversed*), levels (**Auto**, or **Each its own** for one scale per slice), gamma, **Opacity** and **Fade low values**, which makes the weak intensity see-through so the planes behind show.
- **Axes. Draw the axes, with ticks and values**. *Placement* is automatic — on the outer edges facing you, following the view as it turns — or chosen: for each axis, the edge of the cube it runs along and whether it counts **from min** or **from max**.
- **Camera**: azimuth, elevation, depth aspect; **Look into it** faces a chosen plane. **Outline the faces** and an **Orientation tripod** are optional.
- **Save the view...** exports the picture through the same export window as any panel (PNG, TIFF, JPEG at a chosen size and dpi). **To figure composer...** adds the view as a panel of a figure.

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12. Cut arithmetic: dichroism and reference division

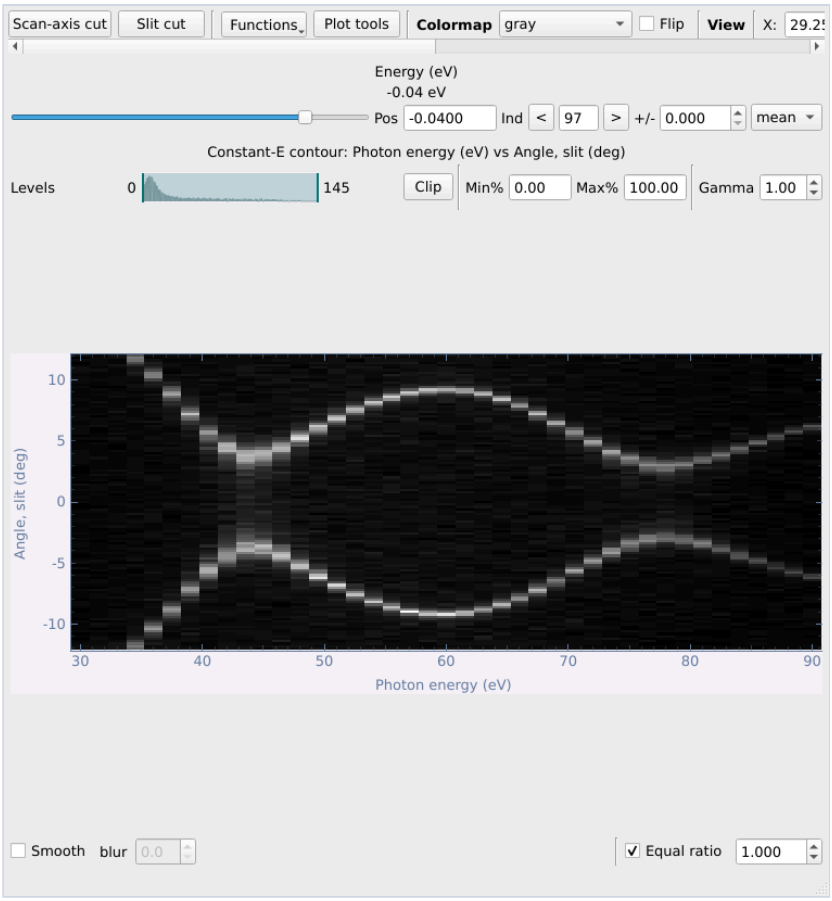
Two cuts combined into a third: linear dichroism (LH − LV), circular dichroism, dividing by a gold reference, or plain differences, ratios and sums. Select two cuts in the list and press tool panel → Advanced → **Cut arithmetic (two cuts)...** (the first selected is A; the window can swap them). The button is greyed until exactly two cuts are selected.



- **The table at the top** compares what the two measurements recorded (polarisation, photon energy, angles, sample position, temperature, pass energy, lens mode). Rows in orange differ beyond a tolerance — a dichroism map between two spots on the sample is a map of *that* difference.
- **Purpose** presets: *Linear dichroism* (A − B, B scaled to the same total intensity), *Circular dichroism* ((A − B)/(A + B)), *Divide by a reference*, *Custom*. Warnings appear if the polarisations look swapped (**Swap A ↔ B**).
- **Scale B to A** by total intensity or by a region; **Hide where A+B below** blanks the background.
- A and B are drawn on one colour scale; the result on a red-white-blue scale symmetric about zero. With raw counts on both, the Poisson uncertainty is propagated and summarised under the plot.
- **Result to list**, **Uncertainty to list**, **All three as a figure**.

13. Photon-energy scans (kz)

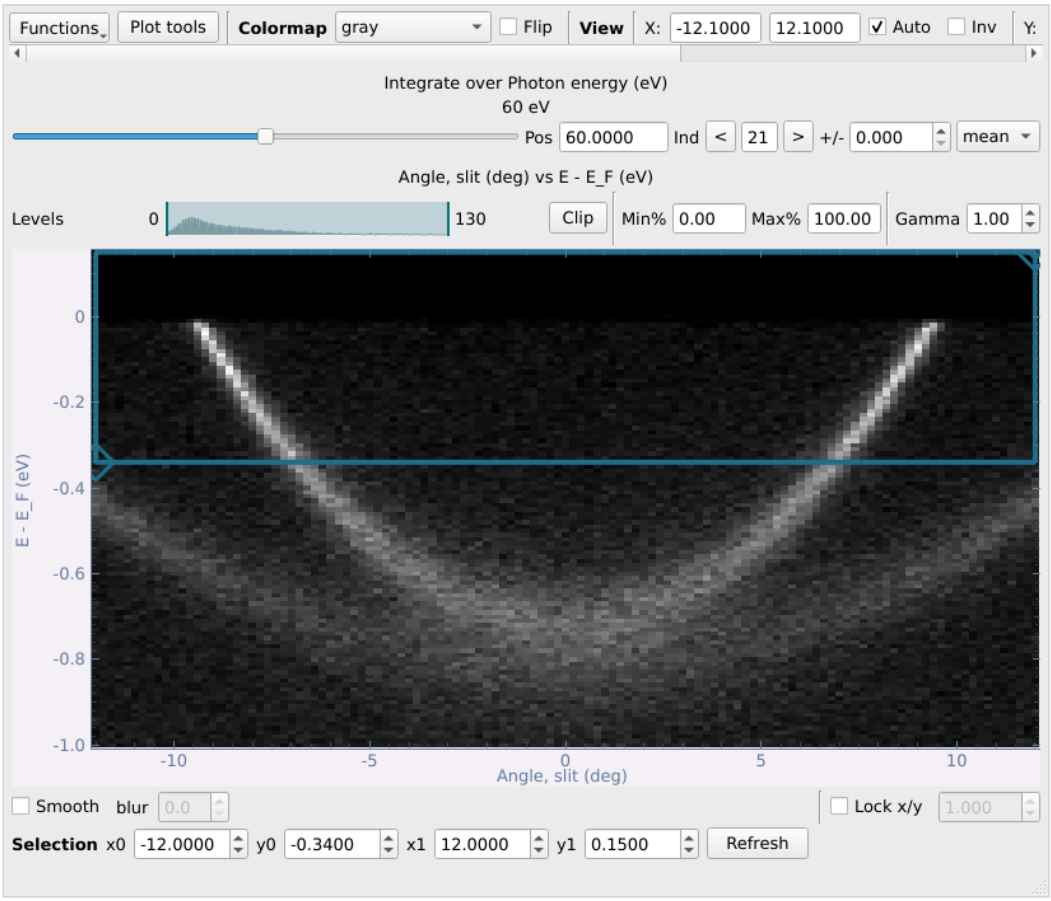
A photon-energy scan is listed as a `kz map`; its contour is photon energy against angle at one energy.



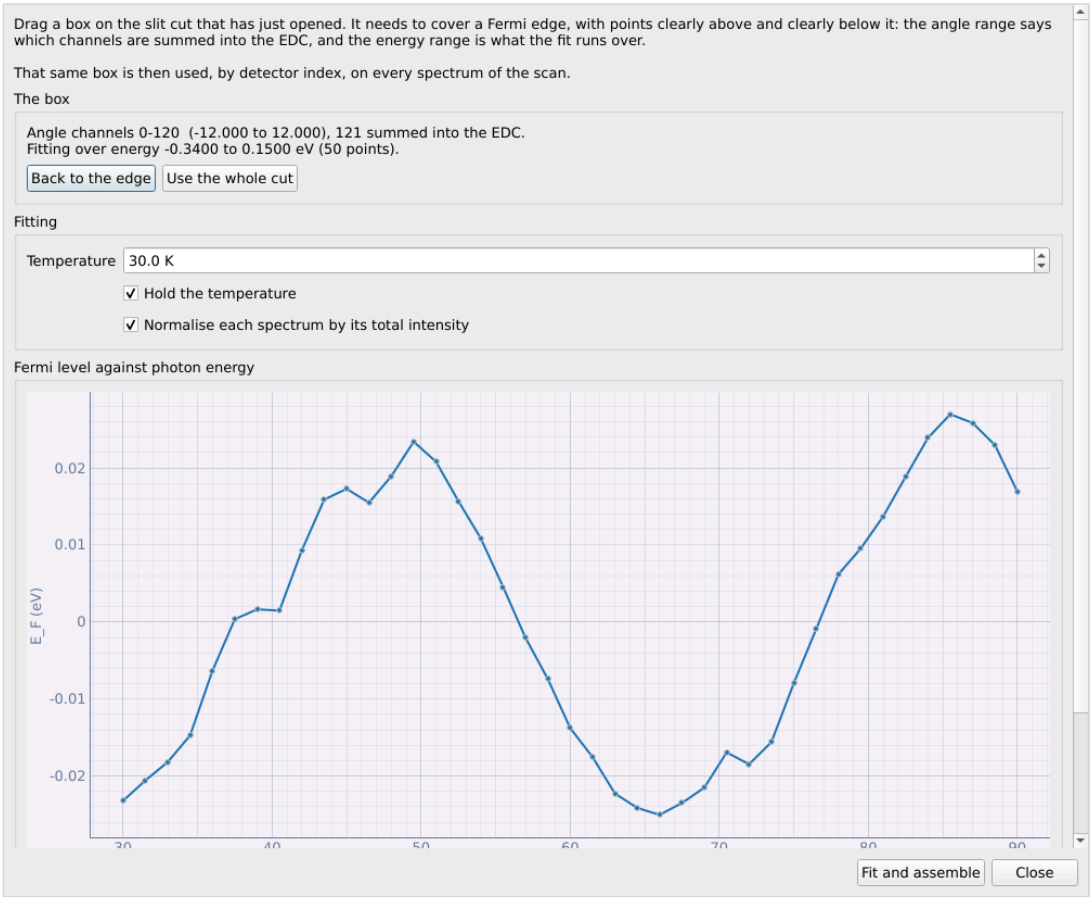
13.1 Aligning the Fermi levels of a kz map

Nothing in the files says where each spectrum's Fermi level really is. Tool panel → Basic → **Align Fermi levels (kz)...** measures it from the spectra themselves; the kz map's slit cut opens with it, and the two close together.

1. The slit cut opens with a box on it. Drag the box over a Fermi edge — the angle range sets the channels summed, the energy range the fit window.



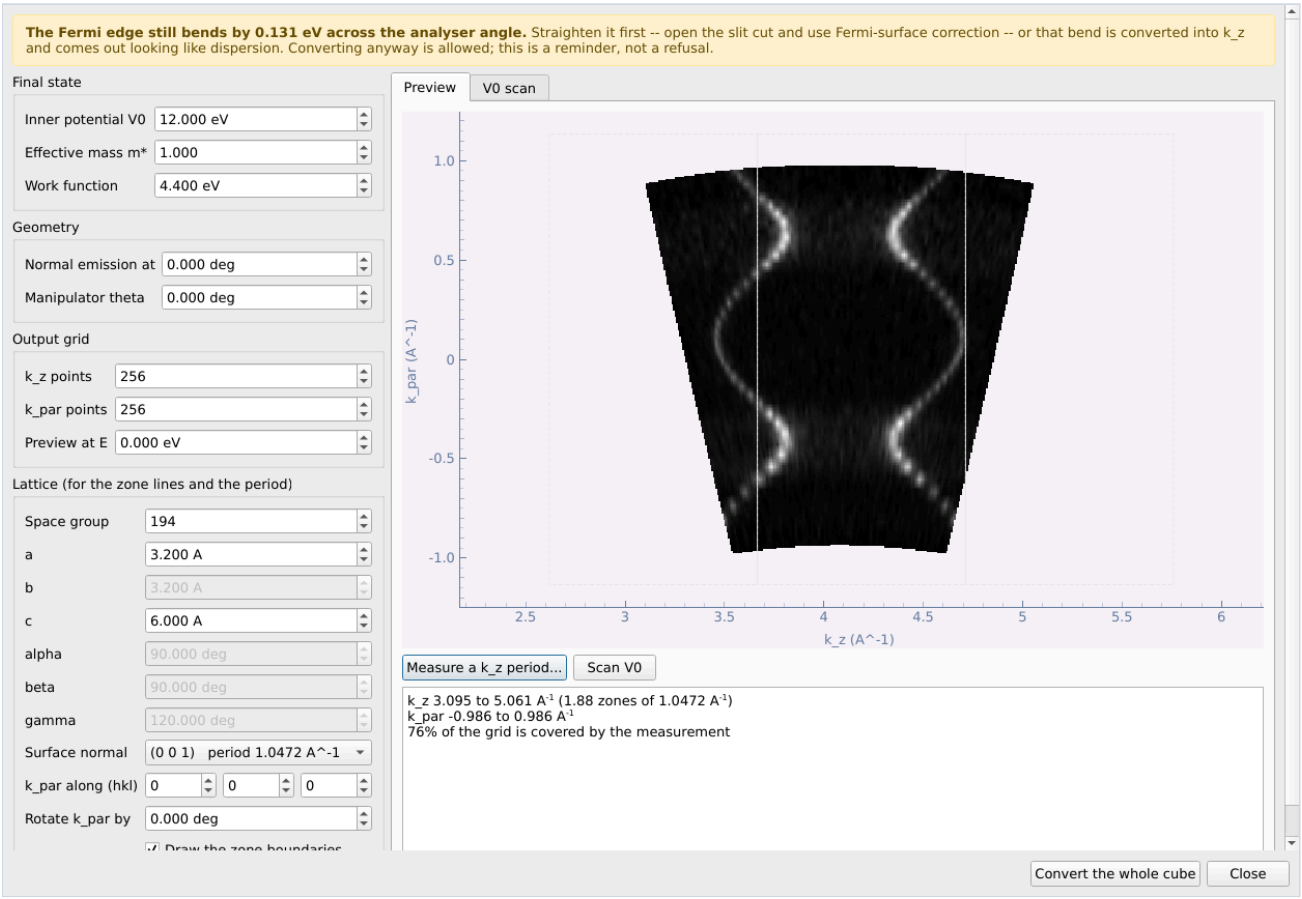
2. **Fit and assemble**: the same box (by detector index) is used for every spectrum; each is shifted to put its own edge at zero; the cube is cropped to the range they all cover; and, by default, each spectrum is **normalised by its total intensity**, which divides out the photon flux.



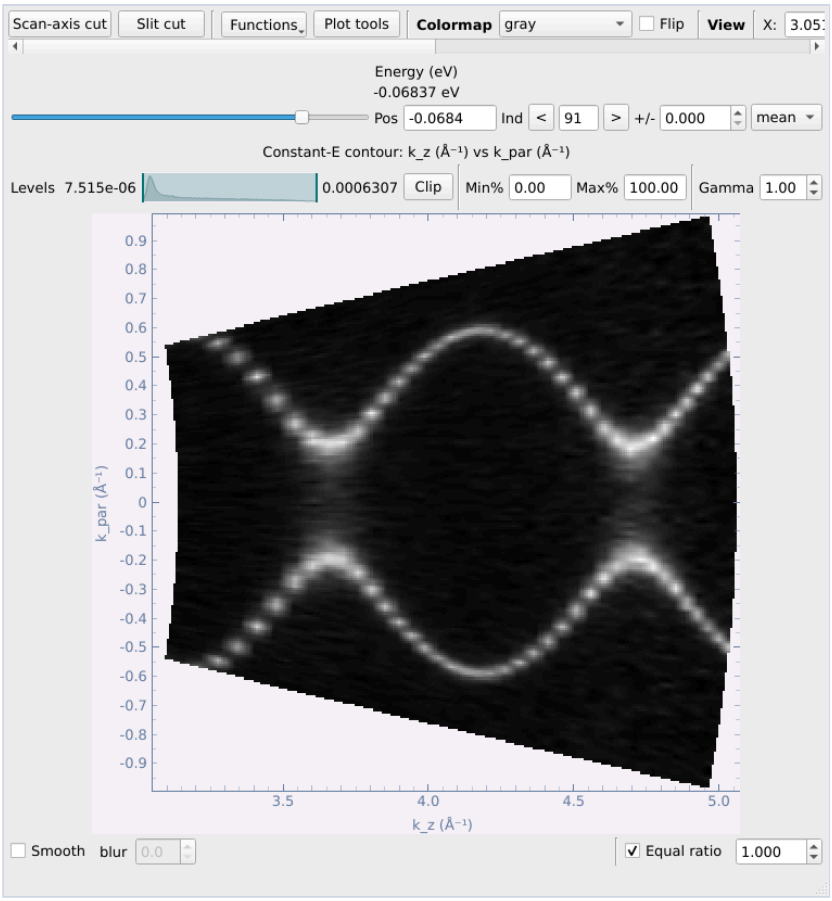
E_F against photon energy is plotted. A smooth curve means the fits are sound; points whose fit is not believed are marked and filled in from their neighbours.

13.2 Converting a kz map to momentum

Tool panel → Basic → **kz conversion...** converts the scan to (k_z, k∥). It starts from the last conversion of a scan of the same size.



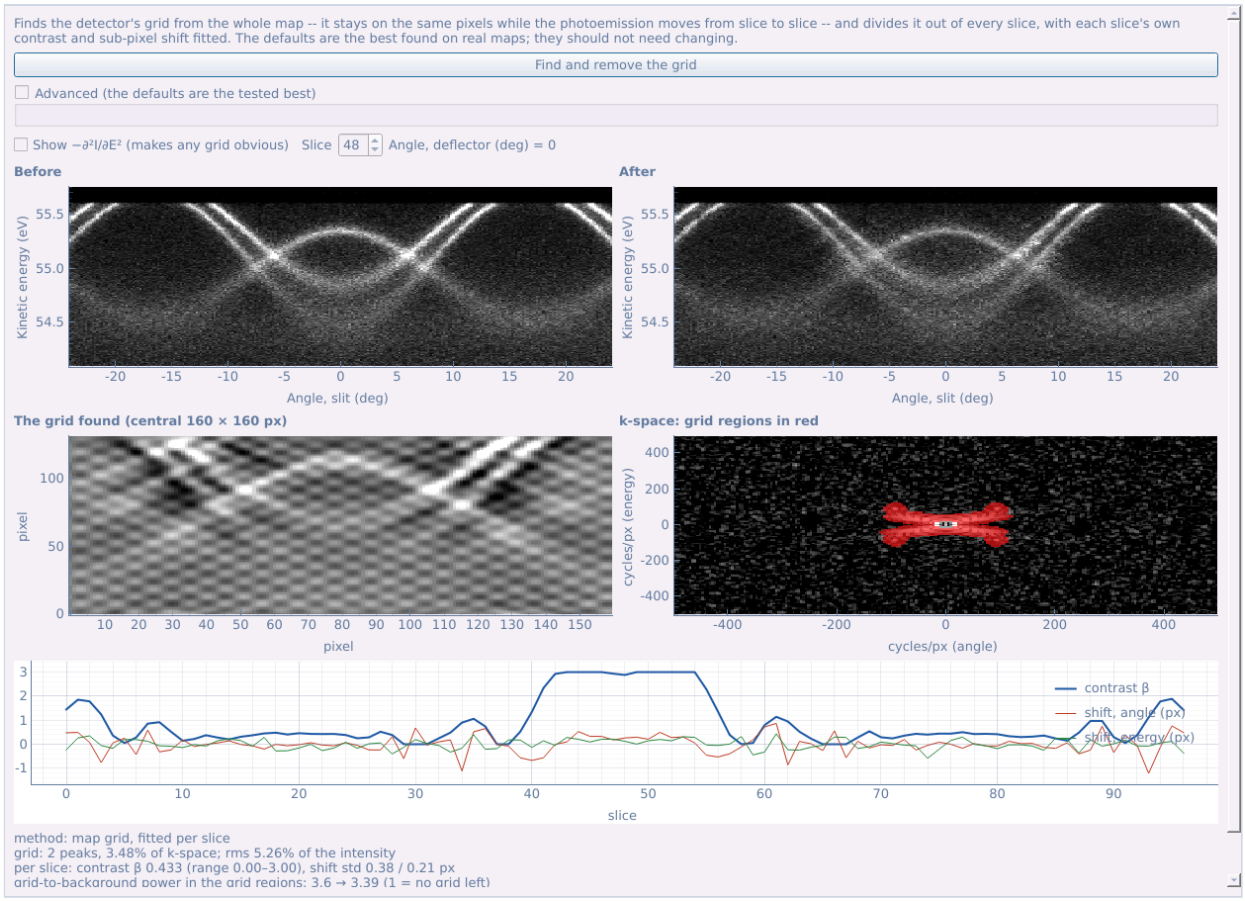
- **Inner potential V0** is the unknown: the preview reconverts one energy slice as you change it, in milliseconds.
- With the **lattice** (space group, a, b, c), the **surface normal** and the in-plane direction, the zone boundaries are drawn over the preview.
- **Scan V0** converts at a range of V0 values and reports where the k_z period matches the lattice — and how well that pins V0.
- **Measure a k_z period...**: click two points that are the same feature one zone apart; the matching reciprocal-lattice planes are listed.
- The yellow banner warns when the Fermi edge still bends across the analyser angle — correct it first (Straighten bands (FS correction) on the slit cut).
- **Convert the whole cube** lists a `kz map (k)`.



14. Removing the detector grid

MCP (hexagonal) and mesh (square) patterns print on every image. **Do it first**: only data still on the detector's pixels can be de-gridded, so the button refuses anything already converted, corrected or smoothed, and says why.

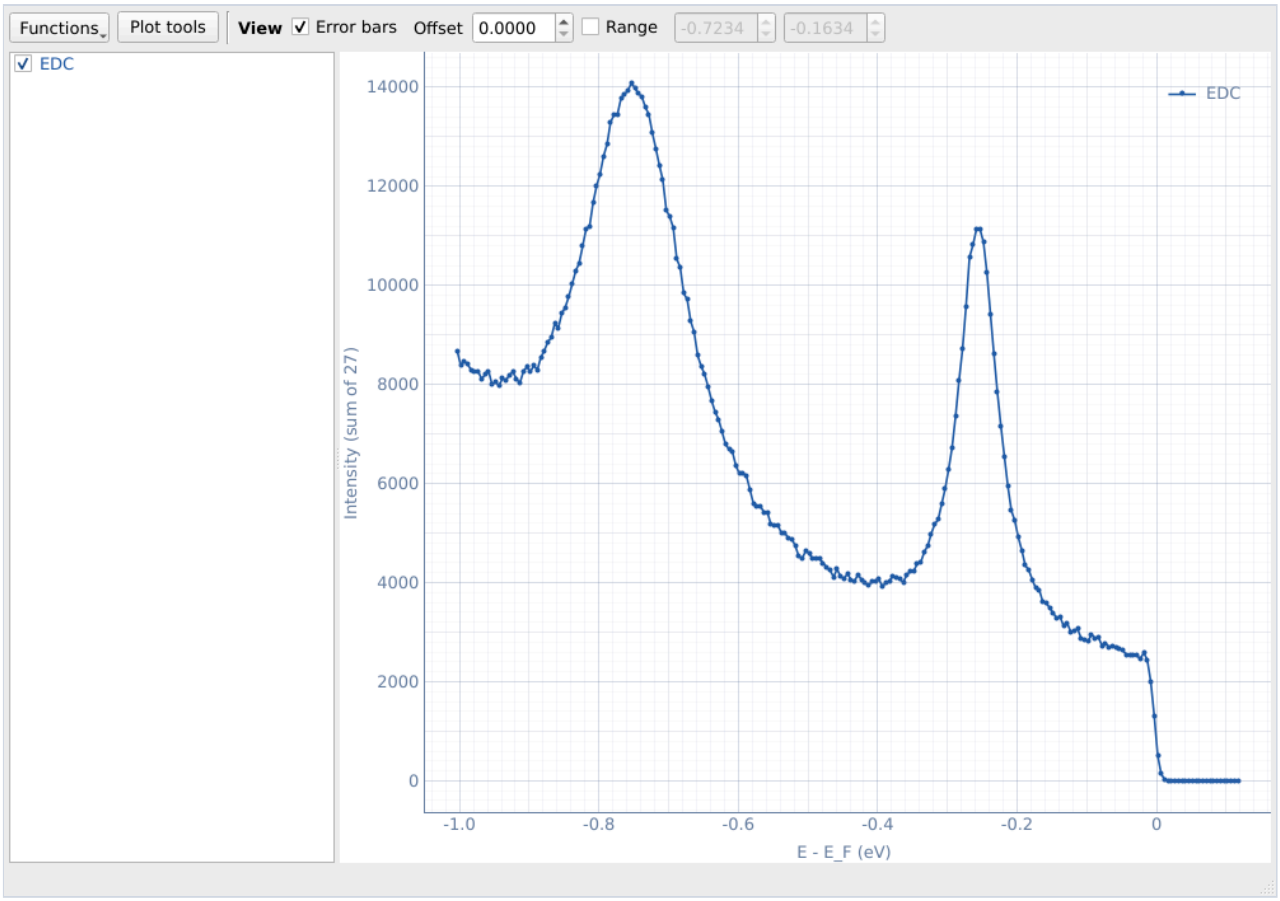
Select a map (or kz map) or a cut and press tool panel → Basic → **Remove detector grid...**. For a map, no reference is needed — the grid stays on the same pixels in every slice while the photoemission moves. **Find and remove the grid**:



- **Before / After** a chosen slice on one colour scale (tick **Show −∂²I/∂E²** to make any grid obvious).
- **The grid found**, its **k-space regions** (in red), and the **contrast and shift** fitted for every slice.
- The power left in the grid regions (1 = no grid left).
- **Also list the grid pattern** keeps a `[grid]` dataset, which a cut taken later with the same lens mode and pass energy will find and use.

15. Curves: EDCs, MDCs and spin-resolved data

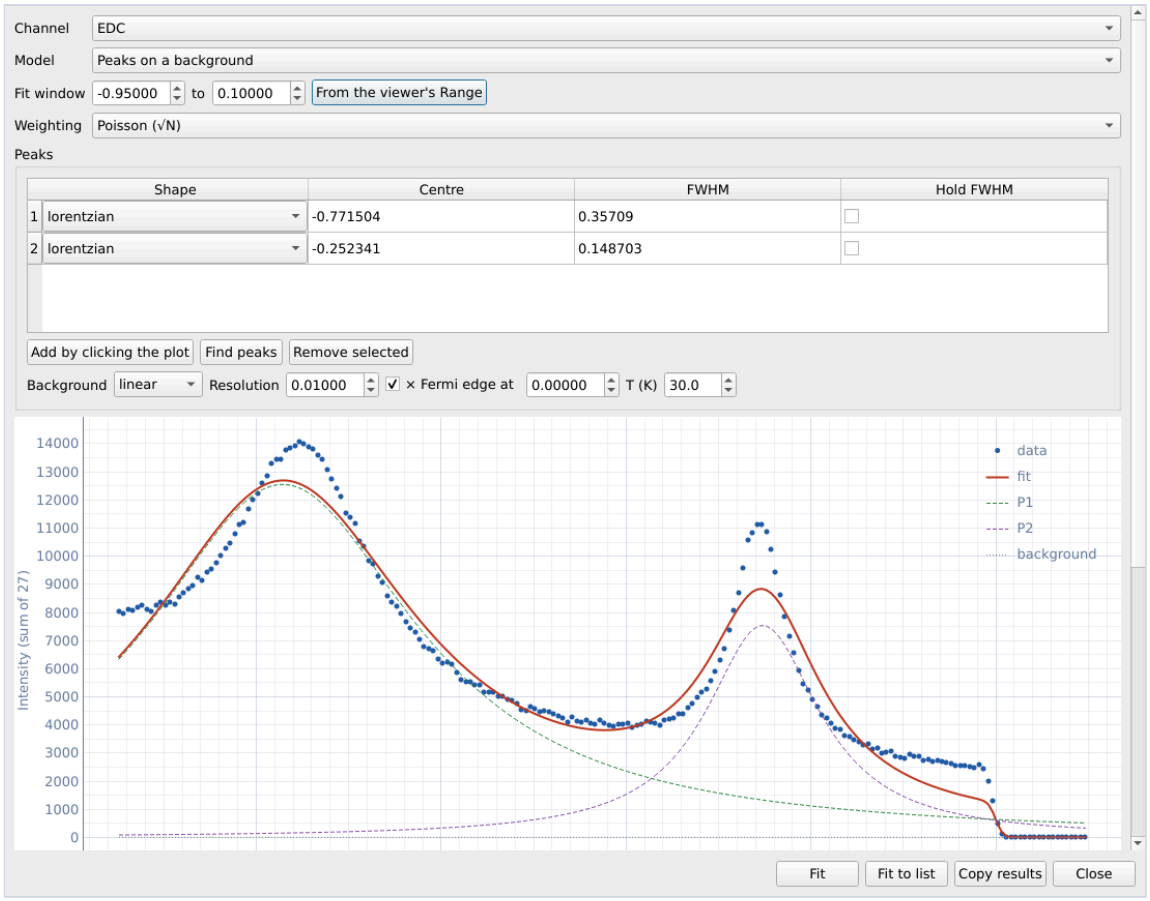
EDC → list / **MDC → list** under any image, or a spin-resolved file, give a curve dataset, which opens in the **curve viewer**:



Channels can be hidden, offset into a waterfall (**Offset**), shown with **Error bars** (from a σ channel), and **Range** marks an x window used by the operations. *Functions → Transform* gives **Crop**, **Bin**, **Normalise**, **Subtract a background** (constant, linear, Shirley), **Shift the axis** and **Add counting errors (√N)**; each makes a new dataset and propagates σ correctly.

15.1 Curve fit

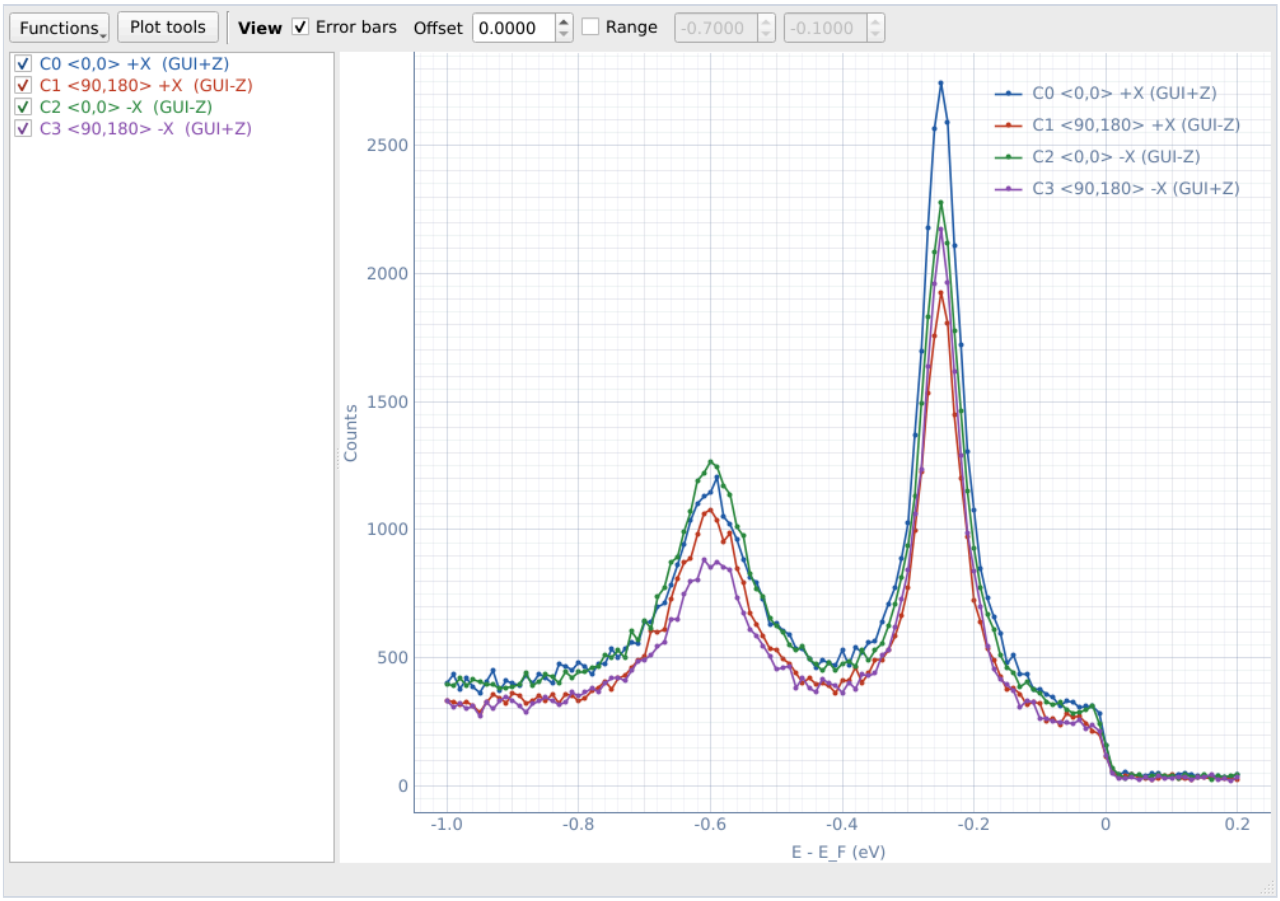
Functions → Curve fit…: **Peaks on a background** (Lorentzian, Gaussian or Voigt peaks placed by clicking or **Find peaks**; none / constant / linear / Shirley background; instrumental resolution; optional **× Fermi edge**), or the **Fermi edge** itself. Weighting is the channel's own σ, √N, or uniform. Every parameter comes with its error; the residual is shown in units of σ.



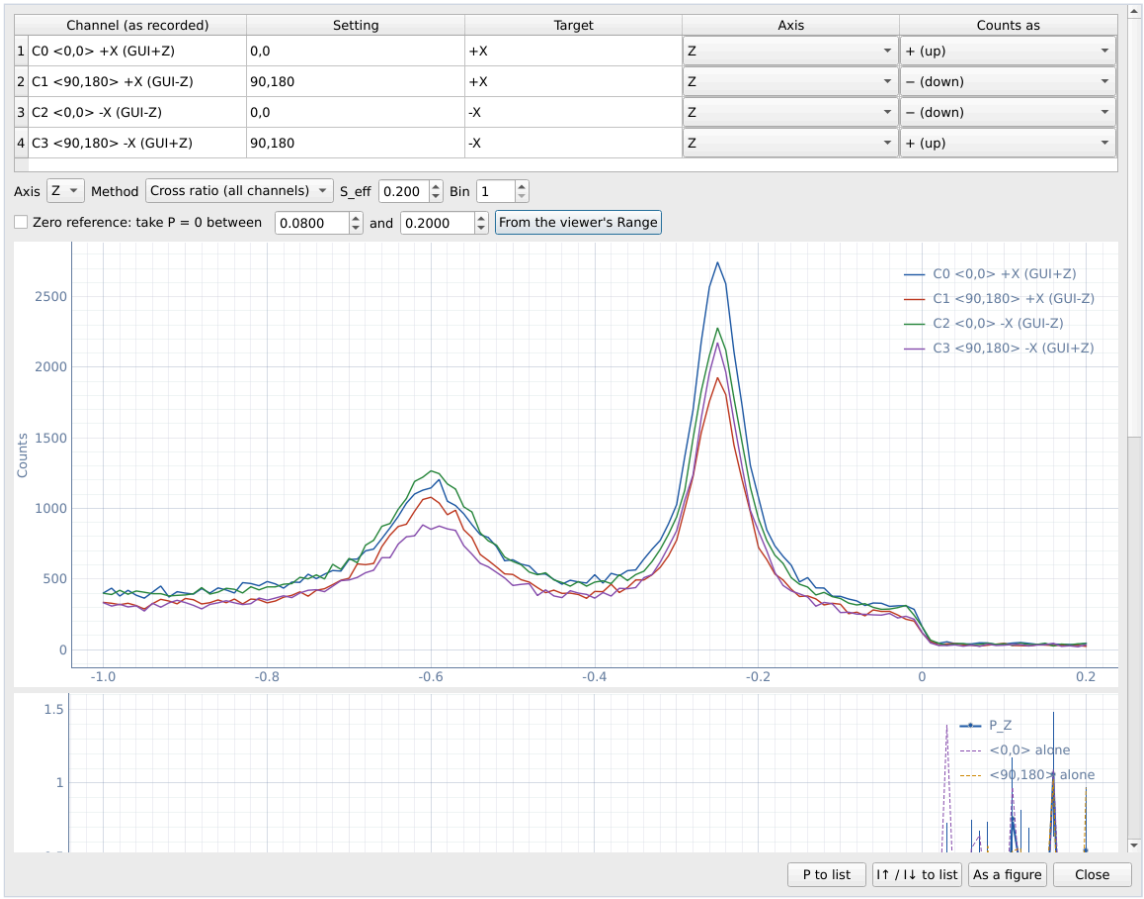
Fit to list keeps data, fit, residual and components; **Copy results** copies a tab-separated table. A Fermi-edge fit can **List a copy with E_F = 0**.

15.2 Spin analysis

A spin EDC opens with one curve per spin channel:



Functions → Spin analysis…:



- The table reads each channel's label (manipulator setting, target magnetisation, axis and sign counted as "up"); **Axis** and **Counts as** can be overridden.

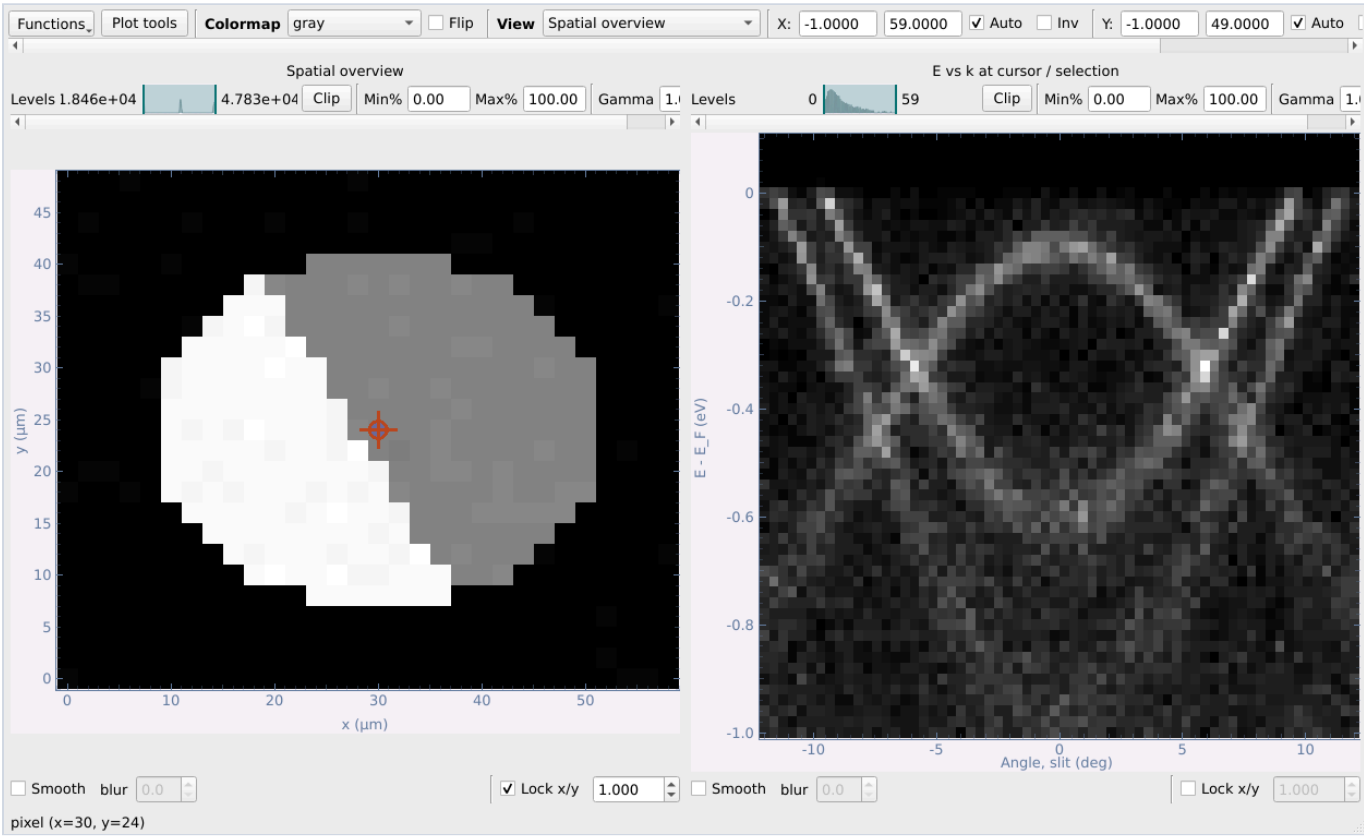
- **Cross ratio** (default when the channels allow it) cancels both the transmission and the target reflectivity exactly; **one pair** is available but carries the instrumental asymmetry, which is measured and reported (ε).
- **S_eff** (Sherman function, default 0.2), **Bin**, optional **Zero reference** window.
- Plots: raw channels, P with its error bars, and I↑ / I↓ with errors.
- **P to list**, **I↑ / I↓ to list**, **As a figure** (opens the figure composer).

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16. Spatially resolved scans (nano-ARPES)

A real-space scan opens the **spatial map** (left, the total intensity at each position) together with the **E vs k spectrum at the cursor** (right). Drag the red target to read the spectrum at another point. Right-click a panel for the two integrations:

- on the map, **Integrate selection → E vs k panel** sums the spectra over a region of the sample;
- on the spectrum, **Integrate selection → spatial panel** maps the intensity of a chosen (k, E) window across the sample — how a domain or a flake edge is found.

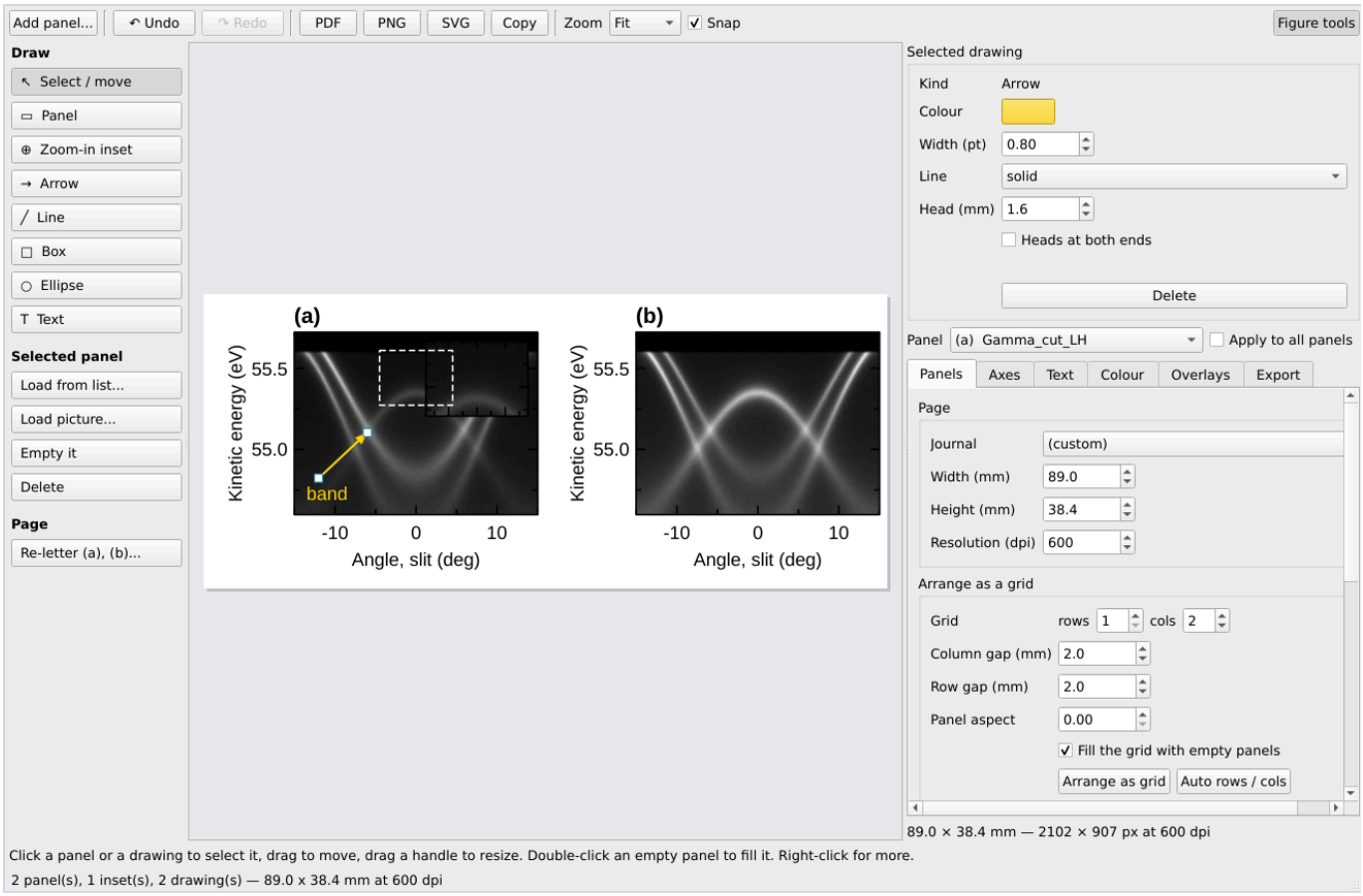


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17. Figures and export

17.1 Figure composer

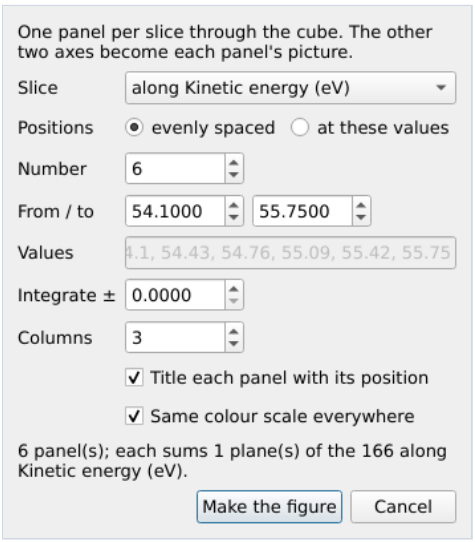
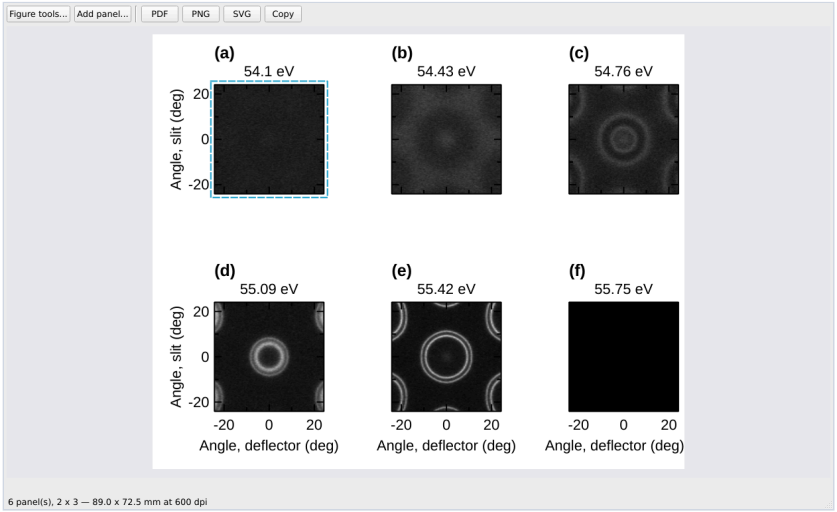
Tool panel → Advanced → **Figure composer...** with **nothing selected** opens a blank page, like a figure of a manuscript: four empty panels (a), (b), (c), (d) at a journal's column width. With cuts (or curves) selected, each becomes a panel (so does a frozen slice's **Figure composer...** button; a stack plot or a 3-D view has its own **To figure composer** button).



- **Lay out the page.** Click a panel to select it; drag it to move it and drag its handles to resize it (Shift on a corner keeps its shape). Panels snap to each other's edges and to a 0.5 mm grid (**Snap**); the arrow keys nudge by 0.5 mm (Shift: 5 mm). *Arrange as grid* (figure tools, Panels tab) puts them back on rows × columns. **Re-letter** names them (a), (b)… in reading order.
- **Fill a panel.** Double-click an empty panel, or use **Load from list...** / **Load picture...** on the left: a dataset from the main list, or a picture from a file (PNG, JPEG, TIFF… — a sketch or a figure made elsewhere; it keeps its shape and has no axes).
- **Insets. Zoom-in inset**: drag out a region of a panel — a small panel magnifying it appears in the corner and the region is outlined. **Panel** drawn inside another panel makes an inset to fill with another dataset. Insets take no letter and move with their panel.
- **Draw. Arrow**, **Line**, **Box**, **Ellipse**, **Text**: drag (or click, for text) on the page. A selected drawing can be moved, its ends or corners dragged, and its colour, width, line style, fill and text set above the figure tools. Double-click a text to edit it; Del removes the selection; right-click for more (bring to front, delete…).
- **Axes and fonts.** The figure tools beside the page act on the selected panel (or all of them): ranges, ticks, labels, which edges carry numbers, the panel's own font size, colour map, colour bar, reference lines.
- **Undo / Redo** (Ctrl+Z / Ctrl+Y) cover every change. **PDF**, **PNG**, **SVG** export, **Copy** puts it on the clipboard; empty panels are never written. The composer's **Figure tools** are not a viewer's **Plot tools**, which draw guides on the live plot.

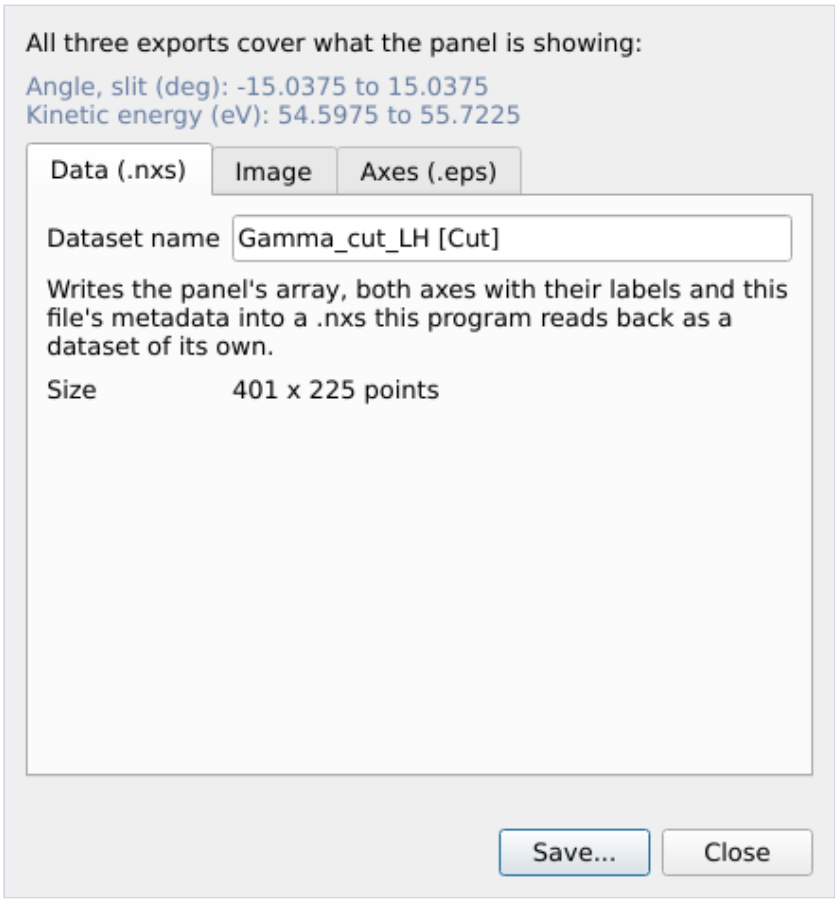
17.2 Slice series figure

Select a map → tool panel → Advanced → **Slice series figure...** lays out a page of flat 2-D slices through the cube as one figure for a paper — constant-energy contours at a series of energies, or cuts at a series of angles or momenta — evenly spaced or at chosen values, with integration, one shared colour scale, and labels on the outer edges only. (Not to be confused with **3D slices**, which draws three planes in a 3-D view.)

Settings	Result
	

17.3 Exporting a panel

Right-click any image → **Export** → **Data (.nxs)...**, **Image...** or **Axes only (.eps)...**. It always exports exactly what the panel shows (range and all): the data as a dataset this program reads back; an image with the colormap applied (PNG, TIFF or JPEG, at a chosen size and dpi, with or without the axes, with the Plot tools guides if there are any, and with the Brillouin zone if the dataset has one); or the frame, ticks and labels as vector PostScript to place the image in, in Illustrator (optionally with the Brillouin zone drawn in the box — see §8.2).



Save datasets... (tool panel, Datasets box) writes the selected datasets to a `.nxs` file of your own; several selected rows go into one file.

18. Autosave, memory and recovery

- **Autosave.** Every computed dataset (a k-map, a processed cut, a fit result…) is written, uncompressed, to this session's folder in `~/.arpes_viewer/sessions/` the moment it exists. Real-space scans are not (they are the measurement itself, and large).
- **A normal close deletes the autosaved copies.** When you close the main window and confirm (everything you want is saved, or *Close anyway*), the open viewers are closed and the session folder is deleted straight away. A dataset's copy also goes as soon as you save it to a file of your own.
- **Recovery after a crash.** Only if the program did not close normally (a crash, a kill) is the session folder kept: the next start offers its datasets back, and it is deleted after three days.
- **Operations log.** `~/.arpes_viewer/sessions/operations.log`, one line per store / save / discard — how to find what was where.
- **Memory box** (bottom of the tool panel). This program's memory, the system's, how many viewers and HDF5 files are open, and how much computed data is held. **Free memory** drops cached copies of computed data (they are re-read from their autosaved file when needed); **Close all viewers** closes every window and then frees memory.
- **Open auto-save folder** (in the Memory box) shows the session folders in the system's file browser, to delete old working copies by hand. The folder of the session now running (named in the status bar) is in use.
- **Long operations** (reading a big map, converting, de-gridding, assembling a folder) run off the GUI thread with a progress bar and a working Cancel.
- **Closing** with unsaved computed data asks first, and names the datasets: **Save to a file...**, **Close anyway** (the autosaved copies are deleted) or **Cancel** (nothing happens).

19. Tips and troubleshooting

- **A greyed tool button** — hover over it: the tooltip says why (e.g. the MDC fit needs a cut already in $Å^{-1}$, cut arithmetic two cuts). Select the right rows in the list and it lights up.
- **A tool is not in a viewer's Functions menu** — it is in the tool panel (see *Where things are now* in §3); check both tabs.
- **A window is taller than the screen** — it opens no taller than the screen, and its contents scroll; the Apply / Close buttons stay at the bottom. Every window can be resized.
- **A k conversion is not offered** — the first axis of the map is not an angle (a photon energy, temperature…). Check *First axis of a map is* in the loader.
- **Cut k conversion of an E − E_F cut** — the energy offset is filled in for you from the Fermi fit; if you shifted the axis some other way, type the kinetic energy of E_F (photon energy − work function) into *Energy offset*.
- **A file is misread** — choose the reader explicitly in the loader, and see *Assumptions worth knowing about* in the README (axis order in SOLEIL files).
- **Dragging the cursor on a big map is sluggish** — a map read lazily from its file re-slices in about 0.1 s per window, so the other windows wait for the mouse to pause. Once the map has been read into memory it follows live.
- **Typing numbers** — every number box takes a decimal point (`0.05`), whatever the system language; a comma is not accepted.
- **The machine is paging** — watch the Memory bar (amber past a third of the RAM, red past a half); use **Free memory** or **Close all viewers**.